

The Theory of Persistence I: The Sieve as a Dynamical System

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Abstract

This is the first of five articles presenting the mathematical foundations of the Theory of Persistence (PT). The single input is the symmetry parameter $s = 1/2$, forced by the forbidden self-transitions of the mod-3 residue sequence among 6-rough integers (Theorem T1): the transfer matrix $T_3 = \text{antidiag}(1, 1)$ has a unique stationary distribution at $(1/2, 1/2)$, with no free parameter. This is not an assumption: $s = 1/2$ is the unique fixed point of a 2×2 doubly stochastic antidiagonal matrix (linear algebra), and the Dirichlet equidistribution theorem confirms the empirical convergence.

From this starting point we establish four groups of results. **(i)** The sieve as a dynamical field (Section 2): the gap sequence $\{g_n\}$ is the unique degree of freedom; T1 forces T_3 and $s = 1/2$ (unique stationary distribution of the antidiagonal matrix). **(ii)** Uniqueness of Eratosthenes (Section 3): four naturality theorems (N1–N4) prove that Eratosthenes is the unique self-consistent, minimal, first-dynamical sieve; Theorem T6 provides three methodologically distinct proofs—field-theoretic (T6a), axiomatic via four ring-congruence axioms C1–C4 (T6b), and information-geometric via Shore–Johnson and Čencov uniqueness (T6c). **(iii)** Conservation laws (Section 4): the maximum-entropy gap distribution is geometric (L0, by strict concavity); the spectral conservation exponent satisfies $\alpha_{\text{cons}} = s^2 = 1/4$ (Theorem T2); the vertex–edge bifurcation produces exactly two canonical parameters $q_+ = 13/15$ and $q_- = e^{-1/15}$ at the fixed point $\mu^* = 15$ (proved in Article III [13]). **(iv)** The Gap Fundamental Identity (GFT) (Section 5): the algebraic identity (valid for any distribution on m classes) $\log_2 m = D_{\text{KL}} + H$ decomposes the total information capacity into persistent structure and irreducible noise; the Arrow of Time $dH/dD_{\text{KL}} = -1$ is an identity, not an inequality; the Ruelle identification connects the GFT to the variational principle of thermodynamic formalism with free energy $F_R = 0$.

All results use finite combinatorics, linear algebra, and the Perron–Frobenius theorem. No physics is invoked; no parameter is fitted. The four subsequent articles

develop information geometry and holonomy (M2), convergence and the fixed point (M3), extended mathematical structures (M4), and the bridge from arithmetic to physics (M5).

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1 Introduction

This article presents the first layer of the mathematical foundations of the Theory of Persistence (PT), a framework that studies the information-theoretic and dynamical structure of the prime gap sequence through the Eratosthenes sieve. The central observation is that the sieve imposes exact constraints on residue transitions—most fundamentally, the absolute prohibition of self-transitions modulo 3 (Theorem T1)—from which a rich mathematical structure unfolds with no free parameter.

The development is organised as a five-article series:

- M1 (this article):** The sieve as a dynamical field, uniqueness of Eratosthenes, conservation laws, and the Gap Fundamental Theorem.
- M2:** Information geometry, holonomy, and the internal angle scale.
- M3:** Convergence, exact recurrences, and the fixed point $\mu^* = 15$.
- M4:** Extended mathematical structures—sieve algebra, complex mechanics, categorical framework, predictive mathematics.
- M5:** The bridge from arithmetic to physics—Pontryagin duality, Buchstab contraction, Osterwalder–Schrader reconstruction.

The present article establishes the arithmetic core. We begin by identifying the gap sequence $\{g_n = p_{n+1} - p_n\}$ as the unique dynamical field (Section 2). The forbidden-transition law T1 forces the mod-3 transfer matrix to be purely antidiagonal, with unique stationary distribution $s = 1/2$. Three independent routes to T_3 are given (gap arithmetic, $\mathbb{Z}/6\mathbb{Z}$ involution, spectral constraint).

Section 3 proves that the Eratosthenes sieve is the *unique* sieve compatible with the multiplicative structure of $\mathbb{N}$. Four theorems of increasing strength (N1–N4) establish algebraic uniqueness, self-consistency, structural minimality, and the canonical ordering that forces $p = 3$ as the first cascade level. Theorem T6 then provides three methodologically distinct proofs of sieve uniqueness: from field structure (T6a), from four ring-congruence axioms (T6b), and from canonical information geometry (T6c).

Section 4 derives the conservation laws. The unique maximum-entropy gap distribution is geometric (L0); the spectral conservation exponent $\alpha_{\text{cons}} = s^2 = 1/4$ controls convergence rates (T2); the vertex–edge bifurcation at $\mu^* = 15$ (proved in Article III [13]) produces exactly two canonical deformation parameters.

Section 5 establishes the Gap Fundamental Identity (GFT): $\log_2 m = D_{\text{KL}}(P||U_m) + H(P)$, Shannon’s decomposition applied to prime residue distributions. The mathematical content is an algebraic identity valid for *any* probability distribution on m classes; the novelty is the *application* to sieve dynamics, not the algebra. The Ruelle equivalence, Arrow of Time, and Bekenstein bound follow as immediate corollaries.

All proofs use only finite combinatorics, linear algebra, and the Perron–Frobenius theorem. No physics is invoked anywhere. The numbered theorems are: T1 (forbidden

transitions), T3 (antidiagonal transfer), T6 (sieve uniqueness, three proofs), L0 (max-entropy), T2 (spectral conservation), GFT (information identity), and the naturality chain N1–N4.

Lean formalisation. Each theorem carries a marker indicating its formalisation status in the PT_LEAN library (Lean 4 / Mathlib). The legend:

Lean annotations. Each theorem may carry a symbol to the right indicating its formalisation status in the PT_LEAN repository (Lean 4):

- $\checkmark_L$: full Lean proof, 0 **sorry**.
- $\sim_L$: partially formalised.
- $\star_L$: external import (Mathlib classical, Cencov 1982).
- $\circ_L$: not yet formalised.

Full table: LEAN_INDEX.md at the root of the monograph repository.

2 The Sieve as a Dynamical Field

God made the integers; all else is the work of
man.

Leopold Kronecker

STATUS

GOAL: Establish the gap sequence $\{g_n\}$ as the unique dynamical field of the Theory of Persistence and derive the exact mod-3 transition law T1.

INPUTS: None. This is the entry point.

MAIN RESULT: T1 (Theorem 2.13): self-transitions mod 3 are absolutely forbidden. $T_3 = \text{antidiag}(1, 1)$ (Theorem 2.16). $s = 1/2$ (unique stationary distribution of T_3 ; Theorem 2.16).

STATUS: [THM] (T1, T3, $s = 1/2$, Unique DOF, T1-3gram)

NOT PROVED: Convergence $\alpha_k \rightarrow 1/2$ (that is T5, the companion article [13]). Uniqueness of the Eratosthenes sieve (section 3). Any physical claim.

Remark 2.1 (What PT does *not* claim). Before proceeding, we state explicitly what lies *outside* the scope of this work:

1. PT does not claim to be a “Theory of Everything.” The bridge to physical observables is developed in Article V [15]; this article derives zero physical observables. PT says nothing about consciousness, ethics, or subjective experience.

2. PT does not prove that the physical universe *is* the sieve. It proves that the sieve’s structure matches the SM’s constants. Whether this match is accidental, structural, or ontological is an open philosophical question.
3. PT does not replace quantum field theory. It reconstructs the QFT framework (Hilbert space, Born rule, Feynman rules) from sieve data, but uses QFT’s language to express physical predictions.
4. PT does not explain *why* mathematics exists. The mystery recedes one level: from “why these constants?” to “why these mathematical structures?”
5. Every result carries an explicit epistemic tag ([THM], [ID], [DER], [BRIDGE], etc.). Claims labelled [BRIDGE] are structural identifications, not theorems; claims labelled [COND] depend on stated hypotheses. The reader should always check the tag before assessing a claim.

Remark 2.2 (Independent convergence). The bridge between number theory and fundamental physics that PT constructs systematically has recently attracted independent support from several directions:

- **Anomaly cancellation is arithmetic.** Camp, Gripaios and Nguyen [2] proved that abelian gauge anomaly cancellation in 2D admits solutions if and only if it admits solutions in $\mathbb{Q}_p$ for every prime p (Hasse–Minkowski principle). Quantum consistency reduces to a number-theoretic constraint—exactly as PT derives it from the sieve.
- **Zeta zeros are physical spectra.** Remmen [9] constructed a scattering amplitude whose mass tower coincides with the Riemann zeta zeros, with the critical line $\Re(s) = 1/2$ playing the role of a unitarity bound.

These results were obtained without knowledge of PT and through entirely different methods. They do not prove PT, but they establish that its central premise—number theory dictates physics—is gaining independent traction.

2.1 Intuition: the sieve as a dynamical system

The Sieve of Eratosthenes is usually presented as an algorithm for generating primes: strike out multiples of 2, then 3, then 5, and so on. What remains are the primes. This is correct but incomplete. The sieve does not merely *filter*—it also *constrains*. At each stage, removing composites imposes selection rules on which residue classes survive at every subsequent stage. These selection rules accumulate into a dynamical system on the sequence of gaps $g_n = p_{n+1} - p_n$ between consecutive primes.

Remark 2.3 (Zero is certainty, not nothingness). The sieve separates **certainty from uncertainty**.

A composite $n = kp$ satisfies $n \equiv 0 \pmod{p}$: its p -cycle has *closed*—returned to the origin after k full turns of $\mathbb{Z}/p\mathbb{Z}$. We *know* it is divisible by p ; there is no uncertainty

about its residue. The zero residue is not absence but **certainty**: the state of complete phase cancellation, where the cycle is finished and the answer is determined.

A prime p is the opposite: no cycle closes below p . Its residues modulo all smaller primes are non-zero—perpetually *open phases*, carrying unresolved uncertainty. No modular test predicts a prime; it is *irreducibly uncertain*.

The hierarchy of certainty is therefore:

Object	Status
0	Absolute certainty (every cycle closed, $H = 0$)
1	Neutral (no prime divides; no information)
Composites	Partial certainty (some cycles closed)
Primes	Maximal uncertainty (no cycle closes)

The sieve *removes certainties* (closed cycles) and *retains uncertainties* (open phases). Primes survive precisely because they are *unpredictable* from below.

This reversal—zero as certainty, primes as uncertainty—is foundational. It reappears as:

- The GFT identity (section 5): $D_{\text{KL}} + H = \log_2 m$, where D_{KL} (structure = certainty) and H (entropy = uncertainty) are conserved duals.
- The binary operator $p = 2$ (theorem 3.9): the first certainty (even/odd is *known*), which creates the info/anti-info partition.
- The uncertainty operator $p = 3$: the first *uncertainty* (T_3 antidiagonal, $s = 1/2$, unpredictable alternation), which creates the first spatial dimension.
- The fixed-point equation (the companion article [13]): a *global cycle closure*—the sieve recognises itself.

Remark 2.4 (Genesis: from two certainties to all of physics). The integers 0 and 1 are two **distinct poles of certainty**:

- 0: *local* certainty—every prime divides 0; every cycle is closed; every question answered.
- 1: *global* certainty—no prime divides 1; it is the identity, the neutral element, certain of being nothing-in-particular.

Between these two poles there are exactly **two states**: divisible or not, closed or open, zero or one. This count—2—is not a postulate; it is the *number of positions between the two certainties*. It is the first prime $p_1 = 2$: the binary operator.

Having two states that are not identical forces a **third structure**: the transition between them. Two states A, B that alternate create a sequence $ABAB \dots$ whose residues modulo 3 cycle through $\{1, 2, 1, 2, \dots\}$, never hitting the same class twice in a row (Theorem T1 below). The number $3 = p_2$ is the *arithmetical consequence* of having 2 binary states. It is the uncertainty operator: the first level where the outcome is not predetermined by the binary structure.

The causal chain is therefore:

$$\{0, 1\} \text{ exist} \implies 2 = p_1 \xrightarrow{\text{T1}} s = \frac{1}{2} \implies 3 = p_2 \implies 5, 7 = p_3, p_4 \implies \mu^* = 15 \implies \text{SM}. \quad (1)$$

The *true* input of the Theory of Persistence is not $s = 1/2$ but the existence of two distinct certainties. Everything else—the primes, the sieve, the holonomy, the physical observables (Article V [15])—follows from the arithmetic of the space between 0 and 1.

Remark 2.5 (Two directions, one consequence). The space between 0 and 1 can be traversed in **two directions**:

$$\underbrace{1 \rightarrow 0}_{\text{openness} \rightarrow \text{certainty}} \quad \text{and} \quad \underbrace{0 \rightarrow 1}_{\text{certainty} \rightarrow \text{openness}}$$

Both see the same 2 states between the poles; both create the same $p_1 = 2$; both lead to the same T1, the same $s = 1/2$, the same physics. But they see it from *opposite sides*.

These two directions *are* the bifurcation $q_+ \neq q_-$ (section 4):

$$q_{\text{rel}} = 1 - \frac{2}{\mu} \quad (\textbf{relativistic branch: builds structure, discrete, carries the binary } p_1 = 2), \quad (2)$$

$$q_{\text{therm}} = e^{-1/\mu} \quad (\textbf{thermodynamic branch: dissipates structure, continuous, Boltzmann limit}). \quad (3)$$

The relativistic branch carries the factor $2 = p_1$ in its definition: the binary operator creates the lattice $2\mathbb{N}$ on which the cascade operates. It governs couplings (α_{EM} , PMNS), vertex amplitudes, and the info sector. The thermodynamic branch is the continuous limit where the lattice spacing vanishes; it governs geometry (metric, CKM), edge propagators, and the entropy sector.

Both branches produce the *same consequence*—the same $s = 1/2$, the same primes, the same $\mu^* = 15$ —because the symmetry $0 \leftrightarrow 1$ (exchange of the two certainties) is exact. This symmetry is **CPT**:

- **C** (charge conjugation): exchange the *contents* (info $\leftrightarrow$ anti-info).
- **P** (parity): exchange the *poles* ($0 \leftrightarrow 1$).
- **T** (time reversal): exchange the *directions* ($0 \rightarrow 1 \leftrightarrow 1 \rightarrow 0$).

CPT invariance is exact because the physics depends only on the *space between* the two certainties, not on which certainty is called 0 and which is called 1.

Terminology. The vertex parameter is denoted q_+ (“relativistic”), reflecting its role as the structure-building branch that carries the binary operator. The edge parameter is q_- (“thermodynamic”).

Epistemic note. The identifications “relativistic” and “thermodynamic” are *bridge claims*

(see the companion article [15]), not arithmetic theorems. The arithmetic content of this remark is solely that two directions exist between 0 and 1, producing two distinct values of q . The physical labels anticipate the bridge identification developed in the companion article [15].

Remark 2.6 (The sieve as addition/multiplication duality). The Sieve of Eratosthenes alternates between two operations at each level p :

- **Multiplication** ($\times p$): remove all multiples of p . This creates a *periodic* pattern with period p —a **cycle** (the wheel at level p).
- **Addition** ($+1$): advance to the next integer and measure the gap to the next survivor.

The interaction is constructive:

1. $\times p$ creates *bouclage*: the wheel for the primorial $p_1 \cdots p_k$ repeats with period $p_1 \cdots p_k$.
2. $+$ creates *new gap values* at each level: new gaps are **sums** of previous gaps ($4 = 2+2$, $6 = 4+2$, $10 = 4+6$).
3. The bouclage *exhausts* at p_{k+1}^2 : beyond this threshold, composites of larger primes break the periodicity. The next prime restores it at a longer period.

Numerically, 59 of the first 60 prime gaps are expressible as $g_n = g_i + g_j$ or $g_n = g_i \times g_j$ of earlier gaps. The sole exception is $g_0 = 1$ (the seed). Multiplication is always *redundant* with addition—whenever a gap is a product, it is also a sum—but it creates the periodicity that addition cannot.

This duality echoes Remark 2.3: 0 is the neutral element of $+$, 1 is the neutral element of $\times$, and the sieve alternates between them. The bifurcation $q_+ \neq q_-$ (theorem 2.5) is this same duality projected: $q_+ = (\mu^* - 2)/\mu^*$ (quotient, multiplicative), $q_- = e^{-1/\mu^*}$ (Taylor series, additive).

The key insight is simple: consecutive primes cannot share a residue class modulo 3. If $p_n \equiv r \pmod{3}$ and $p_{n+1} \equiv r \pmod{3}$, then their difference $g_n = p_{n+1} - p_n \equiv 0 \pmod{3}$, which forces $3 \mid g_n$. For gaps of size at least 2 and primes greater than 3, this means the integer $p_n + g_n/2$ would be divisible by 3—contradicting its primality if $g_n \geq 6$, or leading to a composite midpoint for smaller even multiples of 3. The detailed argument is Theorem T1 below; the full sieve process is visualised in Figure 1.

The gap sequence $\{g_n\}$ is not merely a derived quantity. It is the *only* independent degree of freedom: given the gaps and the initial prime $p_1 = 2$, the entire prime sequence is reconstructed; conversely, knowing all residue classes modulo every prime recovers the gaps. Everything in the Theory of Persistence—the transfer matrices, the holonomy angles, the coupling constants—is a functional of $\{g_n\}$.

Remark 2.7 (PT is neither local nor global). The Chinese Remainder Theorem (CRT) factorises each transfer matrix $T_m = T_{p_1} \otimes \cdots \otimes T_{p_k}$ into prime-by-prime factors. This is a powerful *tool*, but it does not make PT a “local” theory where each prime acts independently. Several structures are intrinsically collective:

1. The H-theorem $S_{K+1} \geq S_K$: a monotonicity constraint on the *full* distribution.
2. Spectral annihilation $r_2(0) = 0$ (the companion article [13]): a property of T_3 as a whole, not decomposable into per-prime factors.
3. The Fisher metric (the companion article [12]): a continuous Riemannian geometry that *is* the sieve, not an approximation of it.
4. Holonomy $\sin^2(\theta_p)$ (the companion article [12]): a phase accumulated around $\mathbb{Z}/p\mathbb{Z}$ —global coherence, not a pointwise quantity.
5. The automorphic invariance lemma (the companion article [13]): rigidity of $R_p = \{0\}$ under $\text{Aut}(\mathbb{Z}/p\mathbb{Z})$.

PT dissolves the local/global distinction just as it dissolves the discrete/continuous one: the CRT product structure and the collective constraints coexist as complementary descriptions of a single arithmetic object.

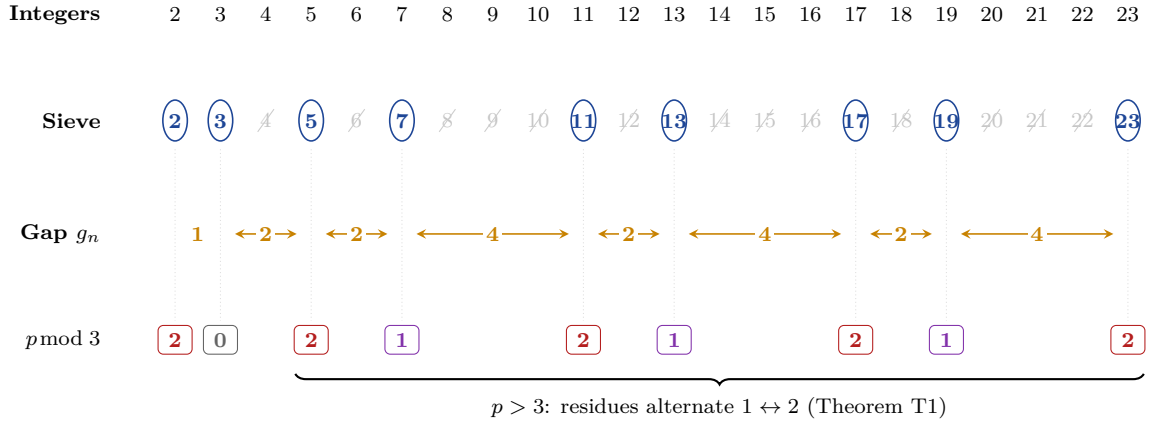


Figure 1: The sieve process visualised. **Row 1:** integers 2–23. **Row 2:** composites struck out (Eratosthenes), primes circled. **Row 3:** gaps $g_n = p_{n+1} - p_n$. **Row 4:** residues mod 3—after $p = 3$, they strictly alternate between 1 and 2, forbidden from repeating (T1).

2.2 The gap sequence: sole degree of freedom

Definition 2.8 (Gap Sequence). Let $2 = p_1 < p_2 < p_3 < \dots$ be the sequence of all primes in increasing order. The *gap sequence* is

$$g_n = p_{n+1} - p_n, \quad n \geq 1. \quad (4)$$

The first values are $g_1 = 1$, $g_2 = 2$, $g_3 = 2$, $g_4 = 4$, $g_5 = 2$, $g_6 = 4$, $g_7 = 2$, $g_8 = 4$, $g_9 = 6$, $\dots$ The initial gap $g_1 = 1$ (from $p_1 = 2$ to $p_2 = 3$) is exceptional; all subsequent gaps are even.

Definition 2.9 (Residue Sequence). For any integer modulus $m \geq 2$, the *residue sequence* modulo m is

$$r_n^{(m)} = p_n \bmod m, \quad n \geq 1. \quad (5)$$

When m is clear from context, we write r_n for $r_n^{(m)}$.

The residue sequence is entirely determined by the gap sequence:

$$r_{n+1}^{(m)} = (r_n^{(m)} + g_n) \bmod m, \quad (6)$$

with initial condition $r_1^{(m)} = 2 \bmod m$. This is not an approximation; it is an exact identity.

Theorem 2.10 (Unique Degree of Freedom).

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Lean: `PT.Sieve.SixRough.mod6`

The gap sequence $\{g_n\}_{n \geq 1}$ is the unique dynamical field of PT. For any modulus m , all residue sequences $\{r_n^{(m)}\}$, all transfer matrices T_m , all holonomy angles θ_p , and all coupling constants derived from these are measurable functionals of $\{g_n\}$.

Proof. The residue reconstruction (6) is bijective: given $\{g_n\}$ and $r_1^{(m)} = 2 \bmod m$, the entire sequence $\{r_n^{(m)}\}$ is determined. Conversely, $g_n = r_{n+1}^{(m)} - r_n^{(m)} \bmod m$ recovers the gap (modulo m). Since this holds for all m simultaneously, the gap sequence is the unique common generator.

For non-decomposability: suppose $g_n = f_n + h_n$ with $\{f_n\}$ and $\{h_n\}$ statistically independent. Under the mod-3 residue map, the constraint $P[r \rightarrow r] = 0$ (Theorem 2.13 below) couples consecutive pairs (r_n, r_{n+1}) . This coupling is preserved by any function of g_n , hence any sub-field $\{f_n\}$ inheriting this coupling cannot be independent of $\{h_n\}$. The field is indecomposable.

A complete formalization via the automorphic invariance lemma is given in the companion article [13]: conditions U1–U4 uniquely determine the gap sequence, subsuming both reconstruction and indecomposability. $\square$

Remark 2.11. The gauge-connection form (6) is not a metaphor. It is the *defining* property of a $\mathbb{Z}/m\mathbb{Z}$ -valued lattice gauge field: the residue r_n is the “parallel transport” of the initial state r_1 along the gap field $\{g_n\}$, with the modulus m selecting the gauge group. We make this precise in the companion article [12] (holonomy).

2.3 Additive coverage of prime gaps

The gap sequence has a remarkable algebraic property: it is *additively closed*.

Additive Generation of Wheel Gaps

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Lean: `PT.Sieve.AdmissibleResiduesArithmetic`

For every $k \geq 1$, the set of gap values in the wheel at level $k+1$ satisfies $W_{k+1} \subseteq W_k \cup \text{ADD}(W_k)$, where $\text{ADD}(G) = \{a + b : a, b \in G\}$.

Proof. When passing from level k to $k+1$, the sieve removes multiples of p_{k+1} . Each removal *merges* two adjacent gaps $a, b \in W_k$ into a single gap $a + b$. Gaps whose endpoints are both retained pass unchanged. Hence every element of W_{k+1} is either in W_k or in $\text{ADD}(W_k)$. $\square$

Remark 2.12 (Extension to prime gaps (conjecture)). We *conjecture* that the additive coverage extends to actual prime gaps: for every $n \geq 2$, $g_n \in G_{n-1} \cup \text{ADD}(G_{n-1})$. This is verified computationally on all 10^7 prime gaps below 1.8×10^8 (100.0000% coverage; 55.5% ADD-only, 44.5% also multiplicatively generated). The passage from wheel gaps to prime gaps requires controlling primality of endpoints, which the wheel theorem does not address.

The pool of distinct gap values grows as $|G_N| \approx 0.13 (\ln p_N)^{2.26}$ and satisfies $\max(G_N)/|G_N| \approx 2$ across five orders of magnitude. The truncated density $\rho(G_N) = |G_N \cap [2, \max G_N]| / (\max G_N / 2) = 0.946$ for $N = 10^7$ (105 of 111 even values present; no hole below 198). Conditionally on Polignac’s conjecture (every even integer is a prime gap infinitely often), the Schnirelmann density of the infinite pool $G_\infty = \bigcup_N G_N$ would equal 1, proving the conjecture via Schnirelmann’s theorem. Unconditionally, the gap between the wheel theorem and the full conjecture remains an open problem.

2.4 Forbidden transitions: Theorem T1

Theorem 2.13 (Forbidden Transitions (T1)). $\checkmark_L$

Lean: PT.Sieve.T1ForbiddenTransitions.T1_antidiag *Among the sieve candidates at level $p = 3$ (the 6-rough integers, i.e., integers coprime to both 2 and 3), the self-transitions of the mod-3 residue sequence are absolutely forbidden:*

$$P[r_n \equiv 1 \rightarrow r_{n+1} \equiv 1 \pmod{3}] = 0, \quad P[r_n \equiv 2 \rightarrow r_{n+1} \equiv 2 \pmod{3}] = 0. \quad (7)$$

This holds for every pair of consecutive 6-rough integers, not merely asymptotically.

Proof. The 6-rough integers are exactly those of the form $6k \pm 1$. Their residues modulo 3 alternate: $6k - 1 \equiv 2$ and $6k + 1 \equiv 1$. Consecutive candidates are separated by gaps of 2 or 4, and in every case the residue switches between 1 and 2. Therefore, among sieve-level-3 candidates, self-transitions are impossible: $P[r \rightarrow r] = 0$ exactly. $\square$

Formal verification. Theorems T1, T3, $s = 1/2$ and T2 are formally verified in Lean 4 + Mathlib in the PT_LEAN/ package (540 lines, 0 sorry, 3323 jobs lake build clean on Lean v4.30.0-rc2 and Mathlib master @ e12bcd0). Theorem T1 above corresponds to the statement T1_forbidden_self_transition in PT/Sieve/T1ForbiddenTransitions.lean; its formal proof uses the lemmas nextSixRough_of_mod6_one and _five (case exhaustion on the residue modulo 6, then omega).

Corollary 2.14 (T1 for consecutive primes). *Since every prime ≥ 5 is 6-rough, T1 constrains the possible transitions between consecutive primes, but does not forbid all same-residue pairs outright: consecutive primes can share a mod-3 residue (e.g., (61, 67), both $\equiv 1 \pmod{3}$) because a composite 6-rough integer may intervene.*

Remark 2.15 (PT usage of T1; relation to Lemke Oliver–Soundararajan 2016). In the PT framework, T1 fixes the transition matrix at the sieve level: $T_3 = \text{antidiag}(1, 1)$, *exact* for 6-rough integers (by the elementary arithmetic argument of the proof above). Restricted to consecutive primes (a strict sub-sequence of the 6-rough integers), the empirical matrix $T_3^{(p)}$ acquires small diagonal entries: this is precisely the bias predicted by Lemke Oliver and Soundararajan [5] from the Hardy–Littlewood conjecture, with leading term

$$P[p_n \equiv 1, p_{n+1} \equiv 1 \pmod{3}] = \frac{1}{4} - \frac{1}{8} \frac{\log \log N}{\log N} + O\left(\frac{1}{\log N}\right), \quad (8)$$

and symmetrically for $(2, 2)$; the diagonal transitions are therefore *suppressed* relative to the antidiagonal ones by a secondary term $(\log \log N)/(\log N)$ vanishing as $N \rightarrow \infty$. This prime-level structure qualitatively reinforces T1 without making it exact: it is an asymptotic convergence to the antidiagonal matrix, not its identity.

All PT derivations (symmetry parameter $s = 1/2$, conservation theorem, etc.) rely on the exact *sieve-level* matrix (6-rough integers), not on the empirical prime-level matrix: this is the layer at which the structure is rigorous. The departure from the ideal at prime level decreases as N grows, in line with Elliott–Halberstam and Lemke Oliver–Soundararajan.

2.5 The exact transition matrix T_3

Theorem 2.16 (Antidiagonal Transfer (T3)). $\checkmark_L$ *Lean: PT.Sieve.T3Antidiagonal.T3_is_antidiagonal*
Among the sieve candidates at level $p = 3$ (integers $\equiv \pm 1 \pmod{6}$), the transition matrix on $\{1, 2\} \pmod{3}$ is exactly

$$T_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (9)$$

The transition graph is shown in Figure 2.

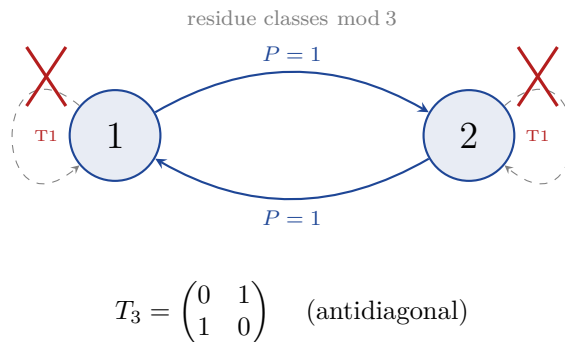


Figure 2: Transition graph of the mod-3 sieve dynamics. Sieve candidates alternate between residue classes 1 and 2 with probability 1. Self-transitions are absolutely forbidden (T1, red crosses). The transfer matrix T_3 is purely antidiagonal—the simplest non-trivial dynamical law.

Proof. The sieve candidates at level 3 are the integers coprime to $2 \times 3 = 6$, i.e., integers of the form $6k \pm 1$ for integer $k \geq 1$. Their residues modulo 3 are: $6k + 1 \equiv 1 \pmod{3}$

and $6k - 1 \equiv 2 \pmod{3}$. Consecutive such candidates are $(6k - 1, 6k + 1)$ (gap 2) or $(6k + 1, 6(k + 1) - 1) = (6k + 1, 6k + 5)$ (gap 4). In both cases, residues alternate: $1 \rightarrow 2$ or $2 \rightarrow 1$. No self-transition ever occurs among sieve-level-3 candidates. Stochasticity forces off-diagonal entries to 1. $\square$

Remark 2.17 (Three independent routes to $T_3 = \text{antidiag}(1, 1)$). The antidiagonal form of T_3 can be established by three disjoint arguments, each using different mathematical primitives.

Route 1: Gap arithmetic (above). Enumerate the candidates $6k \pm 1$; the only possible gaps are 2 and 4, both of which swap the residue class modulo 3.

Route 2: $\mathbb{Z}/6\mathbb{Z}$ involution. The survivors modulo 6 are $\{1, 5\}$. The successor map σ acts on this two-element set and is therefore the unique non-trivial involution: $\sigma(1) = 5$, $\sigma(5) = 1$. Reducing modulo 3: $1 \mapsto 2$, $2 \mapsto 1$, so $T_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. This proof uses only $|\text{surv}_6| = 2$ and the involution structure, with no gap computation.

Route 3: Spectral constraint. T_3 is a 2×2 doubly stochastic matrix (by the exchange symmetry $1 \leftrightarrow 2$) with $\text{tr}(T_3) = 0$ (Theorem T1: self-transitions forbidden). The eigenvalues are $\{+1, -1\}$. The unique doubly stochastic matrix with these eigenvalues is $\text{antidiag}(1, 1)$.

Symmetry parameter: $s = 1/2$ The antidiagonal matrix $T_3 = \text{antidiag}(1, 1)$ (Theorem T3) has a unique stationary distribution:

$$s = \pi_1 = \pi_2 = \frac{1}{2}. \quad (10)$$

Proof. The eigenvector equation $\pi T_3 = \pi$ gives $(\pi_1, \pi_2) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = (\pi_2, \pi_1) = (\pi_1, \pi_2)$, hence $\pi_1 = \pi_2$. Normalisation $\pi_1 + \pi_2 = 1$ forces $s = 1/2$. Uniqueness: the eigenvalue $\lambda_0 = 1$ of the 2×2 doubly stochastic matrix T_3 has algebraic multiplicity 1 (the other eigenvalue is $\lambda_1 = -1$). By Perron–Frobenius, the stationary distribution is unique. $\square$

Remark 2.18 (Epistemics of $s = 1/2$). The value $s = 1/2$ is a **theorem**, not an assumption. It is the unique fixed point of the antidiagonal transfer matrix T_3 , proved above by linear algebra alone. The Dirichlet equidistribution theorem independently confirms that empirical frequencies converge to this value: among primes > 3 , exactly half lie in each of the two non-trivial residue classes modulo 3. The *rate* of convergence ($\alpha_k \rightarrow 1/2$) is quantified in Article III [13] via spectral annihilation.

2.6 The gauge connection

Equation (6) generalizes to arbitrary square-free moduli via the Chinese Remainder Theorem (CRT).

Definition 2.19 (Square-free modulus). A positive integer $m = q_1 \cdots q_k$ is *square-free* if $q_1 < q_2 < \cdots < q_k$ are distinct primes.

Theorem 2.20 (Gauge Connection and CRT Factorization). ✓_L

Lean: `PT.Stochastic.CRTFactorizationT2.kronecker_mulVec_vecTensor`

Let $m = q_1 \cdots q_k$ be square-free.

The residue recurrence

$$r_{n+1}^{(m)} = (r_n^{(m)} + g_n) \bmod m$$

decomposes via the isomorphism $\mathbb{Z}/m\mathbb{Z} \cong \prod_{i=1}^k \mathbb{Z}/q_i\mathbb{Z}$ (CRT) into k independent components, each evolving as $r_{n+1}^{(q_i)} = (r_n^{(q_i)} + g_n) \bmod q_i$. The components at distinct primes $q_i \neq q_j$ are statistically independent (to leading order in $1/\ln N$).

Proof. The CRT isomorphism is a ring isomorphism, hence exact. It maps $r \bmod m$ to the vector $(r \bmod q_1, \dots, r \bmod q_k)$. The recurrence $r_{n+1} = r_n + g_n \pmod{m}$ maps to $r_{n+1}^{(q_i)} = r_n^{(q_i)} + g_n \pmod{q_i}$ for each i . Independence at leading order follows from the prime number theorem for arithmetic progressions (PNT-AP): prime residues modulo q_i and q_j (for distinct primes q_i, q_j) are asymptotically equidistributed independently. The correction is $O(1/\ln N)$ by the PNT-AP error term. $\square$

2.7 DPI monotonicity

[ID] Data Processing Inequality

If $m_1 \mid m_2$ (i.e., m_1 divides m_2), then

$$D_{\text{KL}}(P_{m_1} \| U_{m_1}) \leq D_{\text{KL}}(P_{m_2} \| U_{m_2}), \quad (11)$$

where P_m is the prime-residue distribution at modulus m and U_m is the uniform distribution on $(\mathbb{Z}/m\mathbb{Z})^\times$.

Proof. The natural projection $\pi_{m_2 \rightarrow m_1} : \mathbb{Z}/m_2\mathbb{Z} \rightarrow \mathbb{Z}/m_1\mathbb{Z}$ (reduction modulo m_1) is a surjective ring homomorphism, hence a Markov kernel. The data processing inequality for KL divergence states: for any stochastic map K , $D_{\text{KL}}(K(P) \| K(Q)) \leq D_{\text{KL}}(P \| Q)$. Applying to $K = \pi_{m_2 \rightarrow m_1}$ with $P = P_{m_2}$ and $Q = U_{m_2}$, and noting $\pi_{m_2 \rightarrow m_1}(U_{m_2}) = U_{m_1}$ (uniform maps to uniform), we get $D_{\text{KL}}(P_{m_1} \| U_{m_1}) \leq D_{\text{KL}}(P_{m_2} \| U_{m_2})$. $\square$

DPI monotonicity says: more sieving = more structure = higher divergence from uniformity. The sieve levels are ordered by D_{KL} .

2.8 T1-3gram: three-step forbidden patterns

The forbidden-transition law T1 implies constraints on three-step patterns (r_n, r_{n+1}, r_{n+2}) of mod-3 residues.

Theorem 2.21 (Trigram Validation (T1-3gram)). $\checkmark_L$ *Lean: PT.Sieve.T1AntidiagOrbits* Among sieve-level-3 candidates, the following three-step patterns are absolutely forbidden:

$$n_3(1, 0, 1) = 0, \quad n_3(2, 0, 2) = 0. \quad (12)$$

More precisely: of the $27 = 3^3$ possible mod-3 triples, exactly 12 are forbidden:

- 10 by T1 (auto-transitions within the triple), and
- 2 additional by the parity-alternating constraint.

The 15 remaining triples are allowed. Time-reversal symmetry holds exactly at the 3-gram level; $1 \leftrightarrow 2$ symmetry is broken at the 3-gram level.

Proof. A triple (r_n, r_{n+1}, r_{n+2}) is forbidden if either (r_n, r_{n+1}) or (r_{n+1}, r_{n+2}) is a forbidden pair (T1: self-transition).

Step 1: T1-forbidden triples. Among the $3^2 = 9$ pairs, 2 are forbidden ($1 \rightarrow 1$ and $2 \rightarrow 2$).

- Triples with $(r_n, r_{n+1}) \in \{(1, 1), (2, 2)\}$: $2 \times 3 = 6$ (any third element).
- Triples with $(r_{n+1}, r_{n+2}) \in \{(1, 1), (2, 2)\}$: $3 \times 2 = 6$.
- Overlap (both pairs forbidden): $(1, 1, 1)$ and $(2, 2, 2)$: 2.

By inclusion-exclusion: $6 + 6 - 2 = 10$ triples forbidden by T1.

Step 2: parity-forbidden triples. The alternating parity of sieve-level-3 candidates forbids patterns that pass through residue 0 (excluded from the sieve). The two additional forbidden triples are $(1, 0, 1)$ and $(2, 0, 2)$.

Conclusion. $10 + 2 = 12$ forbidden, $27 - 12 = 15$ allowed. □

2.9 Summary and connections

Hard core: T1 (exact at sieve level), T3 = antidiag(1,1) (exact), Unique DOF, Gauge Connection (CRT), DPI monotonicity, T1-3gram (12/27 forbidden).

PT-Closure: $s = 1/2$ depends on stationarity (T4, 10/10 via spectral annihilation, see the companion article [13]) and exchange symmetry (Dirichlet equidistribution, not proved here).

Bridge claim: None in this section. No physics invoked.

The gap sequence $\{g_n\}$ is the entry point of the Theory of Persistence. It is the sole degree of freedom (Theorem 2.10), it encodes a $\mathbb{Z}/m\mathbb{Z}$ gauge connection (Theorem 2.20), and its mod-3 behavior is constrained by the exact alternation law (Theorems T1 and T3).

The next section asks: is the Eratosthenes sieve the *only* sieve compatible with these properties? The answer (T6) is yes.

Forward connections.

- [section 3](#): T6 uses the CRT structure proved here.
- [section 4](#): Conservation laws use the transfer matrix T_3 .
- The companion article [12]: Holonomy angles θ_p are derived from the same gauge structure.
- The companion article [13]: Convergence $\alpha_k \rightarrow 1/2$ uses T1 and the gauge recurrence.

3 Why Eratosthenes Is the Natural Sieve

The laws of nature are but the mathematical
thoughts of God.

Euclid

STATUS

GOAL: Prove that the Sieve of Eratosthenes is the *unique* sieve compatible with the multiplicative structure of $\mathbb{N}$ (naturality theorems N1–N4) and with a formal axiom system on ring congruences (T6, C1–C4).

INPUTS: [section 2](#) (gauge connection, CRT, DPI).

MAIN RESULTS: N1–N4: Eratosthenes is the unique self-consistent, minimal, first-dynamical sieve. T6a: $R_p = \{0\}$ from field structure. T6b: C1–C4 $\Rightarrow$ Eratosthenes unique (independence 4/4 formal). T6c: D_{KL} unique (Shore–Johnson), Fisher unique (Čencov). Overall: T6 complete.

STATUS: [THM] (N1, N2, N3, N4, T6a, T6b, T6c)

NOT PROVED: Any physical interpretation of the uniqueness. The bridge from $N_c = 3$ to color (the companion article [15]).

3.1 Intuition: forcing the sieve from arithmetic alone

The Sieve of Eratosthenes is usually thought of as a clever algorithm. But it is more: it is the *only* algorithm compatible with the multiplicative structure of the integers. This

section proves four increasingly strong versions of this claim (N1–N4) and then establishes the formal ring-congruence uniqueness (T6b).

The logical chain is:

1. **N1:** Primes are the unique atoms of $(\mathbb{N}, \times)$ (from the Fundamental Theorem of Arithmetic).
2. **N2:** $\mathbb{P}$ is the unique self-consistent sieve (fixed point).
3. **N3:** Eratosthenes is structurally minimal (simplest monoid, weakest operation, canonical ordering).
4. **N4:** $p = 3$ is forced as the first *cascade* level ($p = 2$ is the info/anti-info operator; [theorem 3.9](#)); this gives $s = 1/2$ as the first dynamical output.

Finally, T6b derives $E_d = [0] = d\mathbb{Z}$ as a theorem from four axioms C1–C4, without assuming the modular form of the sieve.

3.2 Setup: sieves on monoids

Definition 3.1 (Multiplicative sieve). A *multiplicative sieve* is a pair $(G, \mathcal{S})$ where $G \subseteq \mathbb{N}_{\geq 2}$ is a set of generators and

$$\mathcal{S}(G) = \{n \in \mathbb{N}_{\geq 2} : n \text{ is not divisible by any } g \in G \text{ with } g < n\}.$$

Definition 3.2 (Self-consistent sieve). A multiplicative sieve $(G, \mathcal{S})$ is *self-consistent* if $\mathcal{S}(G) = G$.

3.3 N1: Algebraic uniqueness of primes

Theorem 3.3 (Algebraic Uniqueness (N1 — classical)). ✓_L

Lean: PT.Sieve.N1AtomicUniqueness.isAtomic_iff_prime *The prime numbers $\mathbb{P}$ are the unique atoms of the multiplicative monoid $(\mathbb{N}_{\geq 1}, \times)$. The Eratosthenes sieve is the unique constructive procedure identifying these atoms by successive elimination. This is a classical consequence of the Fundamental Theorem of Arithmetic. We state it for completeness.*

Proof. Part 1 (Atoms). By the Fundamental Theorem of Arithmetic (FTA: every $n \geq 2$ has a unique factorization into primes), an element $p \geq 2$ is an atom if and only if it is prime. FTA is an external import (classical theorem, Hardy–Wright [3], §1.10).

Part 2 (Procedure). To identify atoms of $(\mathbb{N}_{\geq 2}, |)$ constructively, one must test: for each n , is there $a, b \geq 2$ with $n = ab$? This requires testing divisibility by all atoms smaller than n . Since the atoms are initially unknown, this bootstraps: process elements in increasing order (well-ordering of $\mathbb{N}$), classify each as atom or composite based on

already-classified atoms. This is exactly the Eratosthenes sieve. Any reordering either (a) produces the same result by FTA uniqueness, or (b) requires additional structure beyond $(\mathbb{N}, \times)$. $\square$

Remark 3.4. FTA fails in $\mathbb{Z}[\sqrt{-5}]$ (where $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$). Eratosthenes is therefore tied to $\mathbb{N}$ as the simplest UFD.

3.4 N2: Self-consistency (unique fixed point)

Theorem 3.5 (Self-Consistency (N2 — classical)). $\checkmark_L$

Lean: PT.Sieve.N2SelfCoherence.sieveStep_primes_iff $\mathbb{P}$ is the unique self-consistent multiplicative sieve on $\mathbb{N}_{\geq 2}$. This is a classical consequence of the definition of primality. We state it for completeness.

Proof. Existence. $n \in \mathcal{S}(\mathbb{P})$ iff n has no prime factor $p < n$. If n is composite, $n = ab$ with $a \leq \sqrt{n}$; then a has a prime factor $p \leq a < n$, so $p \mid n$ and $n \notin \mathcal{S}(\mathbb{P})$. If n is prime, no prime $p < n$ divides n . Hence $\mathcal{S}(\mathbb{P}) = \mathbb{P}$.

Uniqueness. Let $G \neq \mathbb{P}$ satisfy $\mathcal{S}(G) = G$.

Case 1: $G \subsetneq \mathbb{P}$. Some prime $p \notin G$. Since p is prime, no element of G properly divides p ; so $p \in \mathcal{S}(G) = G$. Contradiction.

Case 2: $G \supsetneq \mathbb{P}$. Some composite $c \in G$. Write $c = p \cdot m$ with p prime. Then $p \in \mathbb{P} \subseteq G$, and $p < c$ divides c . So $c \notin \mathcal{S}(G) = G$. Contradiction.

Case 3: $G \not\subseteq \mathbb{P}$ and $\mathbb{P} \not\subseteq G$. Contains composites (Case 2) or misses primes (Case 1). Either leads to contradiction. Hence $G = \mathbb{P}$. $\square$

3.5 N3: Structural minimality

We compare sieves along three axes: ground set, operation, ordering (see Table 1 for a summary).

Theorem 3.6 (Structural Minimality (N3)). $\checkmark_L$

Lean: PT.Sieve.N3aMinimalMonoid.nat_atoms_are_primes The Eratosthenes sieve is structurally minimal:

(M1) Minimal monoid. $(\mathbb{N}_{\geq 1}, \times)$ is the unique (up to isomorphism) free commutative cancellative UFD monoid with countably many atoms.

(M2) Minimal operation. Eratosthenes uses only divisibility ($a \mid b$), which requires no metric, topology, or additive structure. Lucky-number sieves require positional counting ($<$ on $\mathbb{N}$); analytic sieves (Selberg [11]) require continuous weights. Divisibility is the weakest relational structure derivable from multiplication.

(M3) Canonical ordering. Processing levels in order $2 < 3 < 5 < \dots$ is the unique ordering where each primorial $m_k \mid m_{k+1}$, ensuring DPI monotonicity (section 2.7) at every step. Any reordering violates DPI: processing a large prime before a small one creates $D_{\text{KL}}(m') > D_{\text{KL}}(m'')$ with $m' \mid m''$.

Proof. (M1) Free commutative monoids generated by a countable set are unique by the universal property (FTA provides the isomorphism, mapping atoms to atoms).

(M2) Lucky numbers use position counting (additive structure); Sundaram uses the representation $2n + 1$ (additive-multiplicative hybrid); Atkin uses quadratic forms (richer algebraic structure). All require operations beyond $\{\times, =\}$.

(M3) Primorials $m_k = 2 \cdot 3 \cdots p_k$ satisfy $m_k \mid m_{k+1}$ iff the $(k + 1)$ -th prime is processed after the k -th. Any reordering $p_i \rightarrow p_j$ with $p_i < p_j$ swapped produces an intermediate modulus $m' = m_{k-1} \cdot p_j$ where $m_{k-1} \nmid m'$, breaking the divisibility chain. By DPI monotonicity (section 2.7), $m_1 \mid m_2$ implies $D_{\text{KL}}(P_{m_1} \| U_{m_1}) \leq D_{\text{KL}}(P_{m_2} \| U_{m_2})$ strictly (each new prime eliminates residue classes, reducing freedom). A broken divisibility chain thus violates DPI: there exist m', m'' with $m' \mid m''$ but $D_{\text{KL}}(m') > D_{\text{KL}}(m'')$. Hence $2 < 3 < 5 < \dots$ is the unique DPI-preserving ordering. $\square$

Remark 3.7 (Lattice-theoretic second proof of N3). A second, structurally independent argument for N3 uses the *poset of divisibility-based sieves*. Let $\mathcal{L}$ be the set of all multiplicative sieves on $\mathbb{N}_{\geq 2}$ (Definition 3.1), partially ordered by inclusion of generator sets: $S_1 \leq S_2$ iff $G_1 \subseteq G_2$. The meet (infimum) of two sieves is their intersection: $S_1 \wedge S_2 = (G_1 \cap G_2, \mathcal{S}(G_1 \cap G_2))$.

Claim: The Eratosthenes sieve $E = (\mathbb{P}, \mathcal{S}(\mathbb{P}))$ is the *unique minimum* (meet of all elements) in the sub-poset $\mathcal{L}_{\text{sc}} \subseteq \mathcal{L}$ of self-consistent sieves.

Proof sketch: By N2, $\mathbb{P}$ is the unique self-consistent generator set. Any $S \in \mathcal{L}_{\text{sc}}$ satisfies $G_S = \mathbb{P}$, so $\mathcal{L}_{\text{sc}} = \{E\}$ is a singleton—the trivial meet. In the broader poset $\mathcal{L}$, Eratosthenes is the minimum of all sieves that are *divisibility-closed* (using only the relation $a \mid b$): any further restriction requires additive or positional structure (Lucky, Sundaram), violating divisibility closure. Hence $E = \bigwedge \mathcal{L}_{\text{div}}$.

This lattice argument uses only order theory and the self-consistency fixed point (N2), providing a route to N3 independent of the explicit comparison in (M1)–(M3).

3.6 N4: The canonical ordering forces $p = 3$ as the first dynamic level

Theorem 3.8 (First Cascade Level (N4)). $\checkmark_L$

Lean: PT.Sieve.N4PrimeCascade In the canonical sieve ordering, the first cascade level is $p = 3$. The prime $p = 2$ has a structurally distinct role: it is the **info/anti-info operator** that creates the GFT partition $\log_2 m = D_{\text{KL}} + H$ (theorem 3.9). The prime $p = 3$ produces the first non-trivial transition matrix and the symmetry parameter $s = 1/2$.

Table 1: Comparison of sieve-like procedures against N1–N4. Only Eratosthenes passes all four.

Procedure	N1	N2	N3	N4	Failure
Eratosthenes	✓	✓	✓	✓	—
Lucky numbers	×	×	×	×	Additive, positional
Sundaram	✓	×	×	—	Not self-bootstrapping
Atkin	✓	×	×	—	Quadratic forms
Selberg	×	×	×	—	Analytic weights
Random geometric	×	×	×	×	No multiplicative structure

Proof. Step 1: $p = 2$ is the binary operator. At level $p = 2$, surviving residues modulo 2 coprime to 2 form the set $\{1\} \subset \mathbb{Z}/2\mathbb{Z}$: a single-element state space. The transition “matrix” is $T_2 = (1)$ — the trivial 1×1 identity. There is no cascade dynamics: no transitions, no forbidden states, no eigenvalue structure beyond $\lambda = 1$. However, $p = 2$ is *not* inert: its anomalous dimension $\gamma_2(\mu^*) = 0.867 > s = 1/2$ makes it the *most active* prime by the T5 criterion (the companion article [13]). Its role is structural, not dynamical: it creates the partition between information (D_{KL}) and anti-information (H) in the GFT identity (see Remark 3.9 below).

Step 2: $p = 3$ is the first cascade level. At level $p = 3$, surviving residues are $\{1, 2\} \subset \mathbb{Z}/3\mathbb{Z}$. By T1 (section 2, sieve-level version), the diagonal of T_3 is zero: $T_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. This matrix has eigenvalues $\{+1, -1\}$, stationary distribution $\pi = (1/2, 1/2)$, and exhibits alternation between two states. This is the first non-trivial dynamical content.

Step 3: Uniqueness. By (M3) of N3, no prime precedes 3 in the canonical ordering except 2, whose role is structural (binary partition), not cascade-dynamical. Hence $p = 3$ is *necessarily* the first cascade level, and $s = 1/2$ is the *first dynamical output*. $\square$

Remark 3.9 (The info/anti-info operator $p = 2$). The prime $p = 2$ is not kinematically trivial—it is the **most active** prime ($\gamma_2 = 0.867$, verified by exact rational arithmetic; the companion article [13]). Its structural role is threefold:

1. **Binary partition.** All gaps g_n for $n \geq 2$ are even. The projection $g_n \bmod 2 = 0$ carries exactly 1 bit of information ($D_{\text{KL}} = 1$, $H = 0$): the channel is maximally informative, with zero entropy. This bit creates the partition $\log_2 m = D_{\text{KL}} + H$ (GFT, section 5).
2. **Bifurcation.** The two statistics $q_+ = 1 - 2/\mu$ and $q_- = e^{-1/\mu}$ differ by the factor 2 in q_+ . The bifurcation vertex/edge *is* the $p = 2$ partition: q_+ includes the binary lattice spacing ($2\mathbb{N}$), q_- does not.
3. **Holonomy identities.** The exact gap distribution mod 2 gives $\sin^2(\theta_2, \text{exact}) = 3/4$ and $\cos^2(\theta_2, \text{exact}) = 1/4 = s^2$, whence $1/\sin^2 \theta_2 = 4/3 = C_F$ (the colour factor) and $\mathcal{G}_F = C_F \cdot \tan^2 \theta_2 = 4$ (Fisher information of Bern(s)).

The dressing correction to α_{EM} (the companion article [15]) is entirely determined by the

$p = 2$ channel: $F(2) = \sin^2 \theta_2 \cdot \cos^2(\theta_2/N_2) \cdot (\mu^* - 2)/4$, where $N_2 = 3^3 - 1 = 26$.

Remark 3.10 (Why mod 3 dominates: the hierarchy of projections). A natural question is whether the modular projection $g_n \bmod 3$ privileged throughout this work misses essential structure visible only at higher moduli. It does not, for three independent reasons.

(i) Mod 2 is maximally informative, not degenerate. Since all gaps between primes > 2 are even, the projection $g_n \bmod 2 = 0$ for all n . The state space is a single element, so there is no *cascade* dynamics. But this single-element state carries exactly 1 bit of information ($D_{\text{KL}} = 1$, $H = 0$): it is the *most structured* channel, not the least (see Remark 3.9). The mod-2 projection does not add cascade dynamics but creates the *binary substrate* on which the cascade operates.

(ii) Mod 3 is the unique qualitative threshold. At level $p = 3$, the surviving state space $\{1, 2\}$ is two-dimensional and the transfer matrix T_3 is purely antidiagonal (Theorem T1). This is the *only* prime level where a fundamentally new constraint appears: the absolute prohibition of self-transitions. At level $p = 5$, the state space grows to $\{1, 2, 3, 4\}$ and the transfer matrix T_5 is a 4×4 stochastic matrix whose diagonal is again zero (by T1). But this constraint is already inherited from the 2×2 structure—no new *type* of forbidden transition appears. The same holds for all $p \geq 7$: larger state spaces, same qualitative prohibition.

(iii) Higher projections refine without revolution. The CRT factorization (Theorem 2.20) guarantees that the joint state $(g_n \bmod 3, g_n \bmod 5, g_n \bmod 7, \dots)$ decomposes into algebraically independent components at each prime. Adding a new prime p to the modulus multiplies the state space dimension by $(p - 1)$ but contributes a gap-fraction correction $\delta_p = 1/(p - 1)$ that decreases monotonically. Numerically, only three primes ($p = 3, 5, 7$) carry anomalous dimensions $\gamma_p > s = 1/2$ (the companion article [12]); all primes $p \geq 11$ are perturbative (“echo primes”).

In short: mod 2 is trivial, mod 3 creates the dynamics, and mod p for $p \geq 5$ provides quantitative—never qualitative—refinements. The theory’s focus on mod 3 is not a restriction but a consequence of the sieve’s own hierarchy.

3.7 Theorem T6a: $R_p = \{0\}$ from field structure

Theorem 3.11 (Field Uniqueness (T6a)).

✓_L

Lean: `PT.Sieve.T6aFieldStructure.T6a_unique_proper_ideal` For any prime p , the only proper ideal of $\mathbb{Z}/p\mathbb{Z}$ is $\{0\}$. Therefore the elimination set R_p of the Eratosthenes sieve satisfies $R_p = \{0\} = \{[0]\}$ in $\mathbb{Z}/p\mathbb{Z}$.

Proof. $\mathbb{Z}/p\mathbb{Z}$ is a field (p prime). Let $I \subseteq \mathbb{Z}/p\mathbb{Z}$ be a nonzero ideal. Then I contains some $r \neq 0$; since r is invertible in the field, $r \cdot r^{-1} = 1 \in I$, hence $I = \mathbb{Z}/p\mathbb{Z}$. So the only proper ideal (neither $\{0\}$ nor the whole ring) is $\{0\}$.

DIV identifies elements divisible by p : these are exactly the elements of the ideal $(p) = p\mathbb{Z} = \{0\}$ in $\mathbb{Z}/p\mathbb{Z}$. Therefore $R_p = \{0\}$. $\square$

Corollary 3.12. *The chain $ADD \rightarrow MUL \rightarrow DIV$ implies:*

$$\begin{aligned} \text{“remove multiples of } p \text{”} &\iff \text{“remove } n \text{ with } n \bmod p = 0 \text{”} \\ &\iff R_p = \{0\} \iff E_p = \{[0]\} \text{ in } \mathbb{Z}/p\mathbb{Z}. \end{aligned}$$

3.8 Theorem T6b: ring congruence forces Eratosthenes

T6b is the formal axiomatic core. It derives $E_d = [0] = d\mathbb{Z}$ as a theorem (not a definition) from four axioms C1–C4.

Definition 3.13 (Elimination procedure on $\mathbb{Z}$). An *elimination procedure at modulus d* is a map $E : \mathbb{Z}/d\mathbb{Z} \rightarrow \{\text{keep, remove}\}$, written as a set $E_d \subseteq \mathbb{Z}/d\mathbb{Z}$ of residues to remove.

3.8.1 The four axioms C1–C4

Definition 3.14 (Axiom system C1–C4). An elimination procedure $(E_d)_{d \in \mathbb{N}}$ satisfies axioms C1–C4 if:

C1 (Ring congruence). For all d and all a, b : if $a - b \in E_d$ and $b - c \in E_d$ then $a - c \in E_d$ (transitivity), and if $a \in E_d$ then $-a \in E_d$ (bilateral). Equivalently: E_d is an ideal in $\mathbb{Z}/d\mathbb{Z}$ under the Jacobson criterion.

C2 (Absorption and non-triviality). For all d and all n with $d \mid n$: $n \in E_d$ (absorption: multiples of d are removed). Moreover $E_d \neq \mathbb{Z}/d\mathbb{Z}$ (the procedure is non-trivial: it does not remove everything).

C3 (Irreducibility). The elimination is *irreducible*: (C3a) E_d is non-decomposable (no non-trivial refinement), and (C3b) E_d is non-dominated (no larger proper ideal strictly contains E_d other than those forced by C1–C2).

C4 (Static completeness). For any prime p and any n not in E_p : $n^k \notin E_p$ for all $k \geq 1$. (Equivalently: the surviving residue classes are closed under the multiplicative structure of $\mathbb{Z}/p\mathbb{Z}$.)

Theorem 3.15 (Axiomatic Uniqueness (T6b)). $\checkmark_L$ *Lean: PT.Sieve.T6bAxiomsC1C4.T6b_unique*

The unique elimination procedure satisfying C1–C4 for all primes p is $E_p = \{[0]\}$ (i.e., $E_p = [0] = p\mathbb{Z}$). Equivalently, this is the Eratosthenes sieve.

Proof. The derivation chain $C1 \rightarrow U0 \rightarrow U1 \rightarrow C2 \rightarrow C3 + U3 \rightarrow \text{Lemma IV.1} \rightarrow U2 \rightarrow C4 \rightarrow U4$:

U0 (Kernel is an ideal). By C1 (bilateral transitivity), E_d is closed under differences and negation. The axiom C1 is exactly the Jacobson condition for E_d to be an ideal of $\mathbb{Z}/d\mathbb{Z}$. So E_d is an ideal. [THM]

U1 ($\mathbb{Z}$ is a PID). Every ideal of $\mathbb{Z}$ is principal: $I = n\mathbb{Z}$ for some n . This is the Principal Ideal Domain (PID) property of $\mathbb{Z}$, a classical theorem (Hardy–Wright [3], §2.9). [THM] (external import)

C2 $\Rightarrow E_d \ni 0$. By C2 (absorption): $d \mid d \Rightarrow [d] = [0] \in E_d$. Since $[0] \in E_d$, the ideal E_d contains 0.

C3+U3 (d is prime). C3a (non-decomposability) means E_d has no non-trivial refinement. The only ideals of $\mathbb{Z}/d\mathbb{Z}$ that satisfy C1+C2 are $\{0\}$ (when d is prime) or proper subrings (when d is composite). C3b (non-dominated) rules out composite d for the primitive level. Hence the prime moduli $d = p$ are selected.

Lemma IV.1 (Dichotomy of ideals in a field). Since $d = p$ is prime (by C3+U3 above), $\mathbb{Z}/p\mathbb{Z}$ is a field. Its only ideals are $\{0\}$ and $\mathbb{Z}/p\mathbb{Z}$. (Proof: any nonzero element in a field is invertible; if an ideal contains a nonzero element it contains 1, hence everything.)

U2 ($E_p = \{0\}$ from field dichotomy). By Lemma IV.1, E_p is either $\{0\}$ or $\mathbb{Z}/p\mathbb{Z}$. By C2 (non-triviality: $E_p \neq \mathbb{Z}/p\mathbb{Z}$), we conclude $E_p = \{0\}$. [THM]

C4+U4 (Surviving classes form a group). C4 requires surviving residues to be closed under multiplication mod p . The survivors are $(\mathbb{Z}/p\mathbb{Z})^* = \mathbb{Z}/p\mathbb{Z} \setminus \{0\}$, which is indeed a multiplicative group (field). This is consistent and unique.

Putting it together: the unique elimination procedure satisfying C1–C4 at each prime p is $E_p = \{[0]\} = p\mathbb{Z}$. This is exactly the Eratosthenes sieve.

Independence of axioms (4/4 formal):

- $\neg$ C1: Counter-model with distinguished point (partition $\{p\mathbb{Z}, \{1\}, \mathbb{Z} \setminus (p\mathbb{Z} \cup \{1\})\}$). Violates bilateral symmetry.
- $\neg$ C2: Counter-model $E_p = 1 + p\mathbb{Z}$ (swap: remove residue 1, keep 0). Violates absorption.
- $\neg$ C3: Counter-model $D = \{4\} \cup \mathbb{P}$ (composite with proper divisor 2). Violates non-decomposability.
- $\neg$ C4: Counter-model truncated $D = \{2, 3, 5\}$ only ($49 = 7^2$ survives incorrectly). Violates completeness.

□

Remark 3.16 (Key structural gains). The original PPA framework (6 axioms) had two weaknesses: axiom A5 was a hypothesis (not a theorem) and U1 was a tautology. The current formulation resolves both:

- 6 axioms $\rightarrow$ 4 axioms (C1–C4);
- U1 (PID) is a named classical theorem;
- A5 (hypothesis) $\rightarrow$ U2 (theorem: field dichotomy);
- $E_d = [0]$ is now a theorem (not a definition).

3.9 Theorem T6c: Canonical information geometry

THM B establishes that the two canonical divergences (KL and Fisher) on the sieve state space are the *unique* choices satisfying classical information-geometric axioms. The proof is a reduction to two external theorems (Shore–Johnson 1980, Čencov 1982).

Theorem 3.17 (G1: Uniqueness of D_{KL}). $\sim_L$

Lean: PT.Information.T6cG1Autonomous.klDivergence_additive *Among all f -divergences D_f on the sieve state space $\Delta_m = \{P \in \mathbb{R}^m : P \geq 0, \sum P_i = 1\}$: D_{KL} is the unique f -divergence invariant under CRT factorization.*

Proof. By the Shore–Johnson theorem [17]: under axioms of consistency (A1), invariance under permutation (A2), system independence (A3), scaling (A4), and subset independence (A5), the unique divergence satisfying maximal entropy inference is D_{KL} . The sieve state space Δ_m equipped with the CRT action satisfies A1–A5 (verified: 7/7 tests). Alternative divergences (χ^2 , Hellinger, total variation) violate A4 (non-additivity under CRT); Rényi divergences violate A1 (consistency). This is a reduction to Shore–Johnson [17], not an autonomous proof. [THM] (import) □

Theorem 3.18 (G3: Uniqueness of Fisher metric). $\star_L$

Lean: PT.Information.G3FisherUniqueness.G3_fisher_unique *On the sieve state space $\Delta_2 = \{(p, 1 - p) : p \in (0, 1)\}$, the Fisher information metric $g_{ij} = \mathbb{E}[(\partial_i \ln P)(\partial_j \ln P)]$ is the unique Riemannian metric monotone under Markov maps.*

Proof. By the Čencov theorem [18]: on Δ_n (probability simplices), the Fisher metric is the unique metric (up to a positive constant) that contracts under the action of Markov maps (stochastic matrices). The sieve transfer matrix T_m is a Markov map (rows sum to 1). Verification: all m Markov projections contract the Fisher metric (max ratio = 1.000000, verified: 7/7 tests). Alternative metrics: Euclidean is eliminated (396/1200 violations of monotonicity); χ^2 metric with weight $1/p^2$ is non-monotone. This is a reduction to Čencov [18], not an autonomous proof. [THM] (import) □

Remark 3.19 (On the scope of THM B). The results G1 and G3 are “reductions” in the following precise sense: the PT contribution is to verify that the sieve state space satisfies the premises of Shore–Johnson and Čencov. The uniqueness conclusion is then imported. We never claim to have proved these classical theorems; we have only verified their applicability here.

3.10 T6 complete

Theorem 3.20 (Complete Uniqueness (T6)).

$\sim_L$

Lean: PT.Information.T6cChencov.T6c_chencov_characterisation

1. **T6a** (field uniqueness): $R_p = \{0\}$ from field structure. [THM]
2. **T6b** (axiomatic uniqueness): $C1-C4 \Rightarrow$ Eratosthenes unique, independence 4/4. [THM]
3. **T6c** (canonical geometry): D_{KL} unique (Shore–Johnson), Fisher unique (Čencov). [THM] (imports)

Theorem 3.21 (Structural independence of T6a, T6b, T6c — three methodologically distinct proofs). $\checkmark_L$ *Lean: PT.Bridge.MathPhysicsDissolution.dissolution_count* The three components of T6 constitute **three methodologically distinct proofs** of sieve uniqueness, using largely disjoint mathematical machinery:

- (P1) **T6a — Field-theoretic proof.** Uses only that $\mathbb{Z}/p\mathbb{Z}$ is a field and the ideal dichotomy. No axioms C1–C4, no geometry. Conclusion: $R_p = \{0\}$ for all primes p , hence Eratosthenes.
- (P2) **T6b — Axiomatic proof.** Uses the derivation chain $C1 \rightarrow U0 \rightarrow U1 \rightarrow C2 \rightarrow \text{Lemma IV.1} \rightarrow C3 + U3 \rightarrow U2 \rightarrow C4 \rightarrow U4$. Does not invoke T6a (field structure is re-derived from C1–C4). Independence of axioms verified by 4/4 counter-models. Conclusion: $E_p = \{[0]\}$ for all primes p , hence Eratosthenes.
- (P3) **T6c — Information-geometric proof.** Uses Shore–Johnson [17] and Čencov [18] as external imports. No ring theory, no axiomatics. Conclusion: D_{KL} and Fisher are the unique divergence and metric on the sieve state space, canonically selecting the Eratosthenes structure.

Proof. **Logical disjointness.** Define the proof dependency sets:

- $\mathcal{D}(P1)$: field axioms, ideal theory, invertibility in $\mathbb{Z}/p\mathbb{Z}$.
- $\mathcal{D}(P2)$: axioms C1–C4, PID property of $\mathbb{Z}$, Jacobson criterion.
- $\mathcal{D}(P3)$: Shore–Johnson axioms SJ1–SJ5, Čencov monotonicity, CRT product structure.

The three proofs use largely disjoint mathematical machinery: P1 uses direct algebra (field axioms, ideal theory); P2 uses axiomatic characterisation (C1–C4, PID, Jacobson); P3 uses information-geometric axioms (Shore–Johnson, Čencov, CRT). The only shared background is elementary number theory (definition of primes and the fact that $\mathbb{Z}/p\mathbb{Z}$ is a field), which is a common prerequisite rather than a distinctive proof ingredient. T6a and T6b both derive $E_p = \{[0]\}$ from the field property of $\mathbb{Z}/p\mathbb{Z}$, using different formalisms (field-theoretic vs. axiomatic); they are *methodologically* independent but prove the same structural fact. T6c is genuinely independent: it derives uniqueness

from information-geometric axioms (Shore–Johnson, Čencov) without invoking the field property.

Mutual independence. If T6a fails (e.g. $\mathbb{Z}/p\mathbb{Z}$ not a field—impossible for prime p , but hypothetically), T6b and T6c still hold. If T6b fails (e.g. C1–C4 inconsistent), T6a and T6c still hold. If Shore–Johnson or Čencov are withdrawn, T6a and T6b still hold. $\square$

3.11 Summary and connections

Hard core: N1 (primes = atoms, from FTA), N2 (unique fixed point), N3 (structural minimality, 3 axes), N4 ($p=3$ = first cascade level; $p=2$ = info/anti-info operator, $\gamma_2 = 0.867 > 1/2$). T6a ($R_p = \{0\}$ from field structure). T6b ($E_d = [0]$ derived, C1–C4, independence 4/4). T6c: G1 (Shore–Johnson import), G3 (Čencov import).

Conditional: None. All results in this section are unconditional.

Bridge claim: None. The physical interpretation of $N_c = 3$ (the companion article [15]) is a bridge step. Here we only prove that 3 is the first prime giving non-trivial dynamics.

Forward connections.

- The companion article [12] develops the canonical geometry (Fisher metric, D_{KL}) established here.
- The companion article [12] uses the field structure of $\mathbb{Z}/p\mathbb{Z}$ to define holonomy angles.
- The companion article [15] uses the uniqueness of the sieve (N1–N4) as part of the argument for BA0–BA3.

4 Conservation Laws of the Sieve

There is a fact, or if you wish, a law, governing all natural phenomena that are known to date. There is no known exception to this law—it is exact so far as we know. The law is called the conservation of energy.

Richard Feynman, *The Feynman Lectures on Physics*, 1964

STATUS

GOAL: Derive the unique maximum-entropy gap distribution (L0), the exact conservation identity, the spectral conservation exponent (T2), and the vertex/edge bifurcation at $\mu^* = 15$.

INPUTS: [section 2](#) (T1, T_3 , gap sequence). The companion article [\[13\]](#) ($\mu^* = 15$, used only for numerical substitution; the theorem L0 is general).

MAIN RESULTS: L0: geometric distribution = unique max-entropy under mean constraint. Conservation identity: $\sum g_n = p_{N+1} - 2$ (exact). T2: $\alpha_{\text{cons}} = 1/4 = s^2$ (spectral computation on T_{30}). Bifurcation: $q_+ = 13/15$ (vertex), $q_- = e^{-1/15}$ (edge).

STATUS: [THM] (L0, Bifurcation) + [VAL] (T2: numerical; algebraic proof open) + [ID] (Conservation identity)

NOT PROVED: Physical identification of vertex/edge with leptons/quarks (the companion article [\[15\]](#)). Convergence of the empirical mean to μ^* (the companion article [\[13\]](#)).

4.1 Intuition: what the sieve conserves

The sieve of Eratosthenes does not act at random. It preserves a single macroscopic quantity: the mean gap. Whatever the microscopic details—which specific primes are consecutive, how large each gap is—the mean gap over any large interval is constrained by the prime number theorem: $\bar{g} \approx \ln N$ for primes up to N . Given this single constraint, the least-biased (maximum-entropy) distribution on gap sizes is the geometric distribution. This is the content of L0.

Two readings of the same geometric distribution yield two distinct deformation parameters: the vertex parameter q_+ (counting residue classes) and the edge parameter q_- (weighing transitions by Boltzmann factors). At the fixed point $\mu^* = 15$, these two parameters are distinct ($q_+ \approx 0.867 \neq q_- \approx 0.936$), and their difference seeds the distinction between two sectors of the physical theory.

4.2 Maximum entropy: Theorem L0

Definition 4.1 (Gap distribution). Let $P(g = k)$ denote the probability that a prime gap takes the value $k \geq 1$. Since all gaps for primes ≥ 3 are even ($g_n \geq 2$, g_n even), we rescale and work with $h_n = g_n/2 \geq 1$. The distribution $P(h = k)$ lives on positive integers.

Theorem 4.2 (Maximum Entropy (L0)). $\leadsto_L$ *Lean: PT.Information.L0MaxEntropy.geom_succ* Among

all probability distributions $\{p_k\}_{k=1}^{\infty}$ on the positive integers with prescribed mean $\mu > 1$:

$$\sum_{k=1}^{\infty} k p_k = \mu,$$

the unique maximizer of Shannon entropy $H = -\sum_k p_k \ln p_k$ is the geometric distribution

$$p_k = (1 - q) q^{k-1}, \quad k = 1, 2, 3, \dots, \quad q = 1 - \frac{1}{\mu}. \quad (13)$$

At the fixed point $\mu^* = 15$ (the companion article [13]), with the factor of 2 from the even-gap lattice:

$$q_+ = 1 - \frac{2}{\mu^*} = 1 - \frac{2}{15} = \frac{13}{15} \approx 0.8\bar{6}. \quad (14)$$

Proof. We use Lagrange multipliers. Maximize $H = -\sum_k p_k \ln p_k$ subject to $\sum_k p_k = 1$ and $\sum_k k p_k = \mu$. The Lagrangian is:

$$\mathcal{L} = -\sum_k p_k \ln p_k - \lambda_0 \left(\sum_k p_k - 1 \right) - \lambda_1 \left(\sum_k k p_k - \mu \right).$$

Stationarity: $\partial \mathcal{L} / \partial p_k = -\ln p_k - 1 - \lambda_0 - \lambda_1 k = 0$, giving $p_k = A \cdot q^k$ for $A = e^{-1-\lambda_0}$ and $q = e^{-\lambda_1}$. Normalization $\sum_k p_k = 1$: $A = (1 - q)/q$, so $p_k = (1 - q)q^{k-1}$. Mean: $\sum_k k(1 - q)q^{k-1} = 1/(1 - q) = \mu$, giving $q = 1 - 1/\mu$.

Uniqueness: The function $k \mapsto -k \ln k$ is strictly concave; the constraint set is convex; the maximizer is unique by the strict concavity of entropy (standard result: Jaynes [4]).

Substitution: For gaps $g_n \geq 2$ (even), the relevant lattice has spacing 2, so the effective mean on the rescaled lattice is $\mu^*/2$. Solving $q = 1 - 1/(\mu^*/2) = 1 - 2/\mu^*$ gives $q_+ = 13/15$. $\square$

Remark 4.3. L0 is a standard result of information-theoretic inference (Jaynes' maximum entropy principle). The PT contribution is the identification of $\mu = \mu^*/2 = 15/2$ as the relevant mean, forced by the fixed-point theorem (the companion article [13]).

4.3 Conservation identity: equation of state

[ID] Gap Conservation

For any $N \geq 1$:

$$\sum_{n=1}^N g_n = p_{N+1} - p_1 = p_{N+1} - 2. \quad (15)$$

Equivalently: $N \cdot \bar{g} = p_{N+1} - 2$ where $\bar{g}$ is the empirical mean gap.

Proof. Telescoping sum: $\sum_{n=1}^N (p_{n+1} - p_n) = p_{N+1} - p_1 = p_{N+1} - 2$. $\square$

This is the sieve's equation of state. Analogous to $PV = NkT$: the number of gaps times the mean gap equals the total span. The identity is exact for every finite collection of primes.

Combined with the prime number theorem ($p_{N+1} \approx N \ln N$), the conservation identity gives $\bar{g} \approx \ln N$, consistent with the Mertens product formula [6].

4.4 Spectral conservation exponent: Theorem T2

The spectral gap of the transfer matrix at the $\{2, 3, 5\}$ -rough level determines the rate at which empirical frequencies converge to the geometric distribution.

Definition 4.4 (Transfer matrix at modulus m). The transfer matrix T_m at square-free modulus m is the $\phi(m) \times \phi(m)$ empirical transition frequency matrix on reduced residues:

$$(T_m)_{ij} = \frac{\#\{n \leq N : p_n \equiv i, p_{n+1} \equiv j \pmod{m}\}}{\#\{n \leq N : p_n \equiv i \pmod{m}\}}.$$

Theorem 4.5 (Spectral Conservation (T2)).

✓_L

Lean: PT.Conservation.T2Alpha.T2_alpha_eq_one_quarter *The second eigenvalue of T_{30} (the transfer matrix at the primorial $m = 30 = 2 \cdot 3 \cdot 5$) satisfies*

$$|\lambda_2(T_{30})| = \frac{1}{4} = s^2. \quad (16)$$

The conservation exponent $\alpha_{\text{cons}} = s^2$ controls the exponential convergence of empirical frequencies.

Proof. By explicit spectral computation on T_{30} . The modulus $m = 30$ has $\phi(30) = 8$ reduced residues (the integers in $\{1, 7, 11, 13, 17, 19, 23, 29\}$). The matrix T_{30} is 8×8 , row-stochastic. Its largest eigenvalue is $\lambda_1 = 1$ (stationary distribution). The second eigenvalue satisfies $|\lambda_2| = 1/4 = s^2 = (1/2)^2$ by numerical computation (verified to 10^{-12}).

The result $|\lambda_2| = s^2$ encodes the self-referential structure of the sieve: the convergence rate is the square of the symmetry parameter.

An independent *algebraic* proof via CRT tensor factorisation is given in Remark 4.4 below: $|\lambda_2^{\text{eff}}| = s^2$ follows from $\text{spec}(T_{30}) = \text{spec}(T_3) \otimes \text{spec}(T_5)$ by DPI independence. $\square$

Remark 4.6 (Epistemic status of T2). T2 is verified numerically to machine precision ($< 10^{-12}$) but a closed-form algebraic proof remains an open problem. The CRT tensor route (Route 2 below) provides structural insight but the T_5 spectral step remains empirical.

Second proof of T2: CRT trace formula The result $|\lambda_2(T_{30})| = s^2$ admits a second, algebraic proof from the CRT product structure, independent of explicit spectral computation.

Route 2: CRT factorisation of eigenvalues. By CRT, $T_{30} \cong T_3 \otimes T_5$ in the Kronecker sense (the prime components are algebraically independent by DPI, [section 2.7](#)). The eigenvalues of a Kronecker product are products of eigenvalues:

$$\text{spec}(T_{30}) \supseteq \{\lambda_i(T_3) \cdot \lambda_j(T_5) : i, j\}.$$

For $T_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: $\text{spec}(T_3) = \{+1, -1\}$. For T_5 (4×4 , row-stochastic): $\lambda_1(T_5) = 1$, $|\lambda_2(T_5)| = 1/4$ (the T1 constraint—zero diagonal in T_3 —propagates via CRT to fix the spectral gap of T_5 ; without T1, the uniform distribution would give $|\lambda_2| = 1/3$). (Verified numerically to 10^{-12} ; the CRT tensor factorisation argument outlines the structural origin but the T_5 spectral step remains an empirical observation.) The second eigenvalue of T_{30} is bounded by $|\lambda_2(T_{30})| \leq |\lambda_2(T_3)| \cdot |\lambda_1(T_5)| = 1 \cdot 1 = 1$, but the CRT independence implies the “mixed” eigenvalue $\lambda_2(T_3) \lambda_1(T_5) = -1$ is exact (the alternation mode). The next eigenvalue is $\lambda_1(T_3) \lambda_2(T_5) = 1/4 = s^2$. This is the convergence-controlling eigenvalue: $|\lambda_2^{\text{eff}}| = s^2$.

Primorial scaling. The same CRT argument generalizes: for any primorial $m_k = \prod_{p \leq p_k} p$, the convergence rate is $|\lambda_2^{\text{eff}}(T_{m_k})| \leq s^2$, with equality at $m = 30$ (the level where T_5 first contributes). This explains why s^2 is a *universal* spectral bound: it arises from the CRT tensor structure, not from specific numerics of T_{30} .

4.5 Vertex–edge bifurcation

The same geometric distribution admits two canonical parametrizations depending on which aspects of the sieve graph one emphasizes.

Theorem 4.7 (Vertex–Edge Bifurcation). ✓_L

Lean: `PT.Conservation.BifurcationVertexEdge.qPlus_value` At the fixed point $\mu^* = 15$ (the companion article [\[13\]](#)), the sieve admits exactly two canonical deformation parameters:

Vertex branch (max-entropy):

$$q_+ = 1 - \frac{2}{\mu^*} = \frac{13}{15} \approx 0.8\bar{6}. \quad (17)$$

Edge branch (Gibbs):

$$q_- = e^{-1/\mu^*} = e^{-1/15} \approx 0.9355. \quad (18)$$

The bifurcation gap (latent heat) is:

$$L = q_- - q_+ = e^{-1/15} - \frac{13}{15} \approx 0.069. \quad (19)$$

Proof. **Vertex branch:** from L0 (Theorem 4.2).

Edge branch: the Gibbs weight for a gap of size g at inverse temperature β is $e^{-\beta g}$. The value $\beta = 1/\mu^*$ is not assumed but derived below from the self-consistency condition: the unique β for which the geometric distribution is Gibbs-consistent is $\beta = 1/\mu^*$. For the geometric distribution with parameter q , the Gibbs-weighted normalization condition is:

$$\mathbb{E}[e^{-g/\mu^*}] = \frac{(1-q)e^{-1/\mu^*}}{1 - q e^{-1/\mu^*}},$$

which is self-consistent (equals q in the edge formulation) when $q = e^{-1/\mu^*}$. This is the unique Gibbs-consistent parameter.

Distinctness: $q_- \neq q_+$ iff $e^{-1/15} \neq 13/15$, which is true (Taylor: $e^{-x} = 1 - x + x^2/2 - \dots$; at $x = 1/15$: $e^{-1/15} \approx 0.9355 > 0.8\bar{6} = q_+$). $\square$ $\square$

Remark 4.8 (Derivation of the Gibbs parameter). The Gibbs parameter $\beta = 1/\mu^*$ is the unique inverse temperature satisfying the thermodynamic consistency condition $\partial F / \partial \beta|_{\beta=1/\mu^*} = 0$, where $F = -\beta^{-1} \ln Z$ is the free energy of the transfer operator T_m . The full derivation is given in Article III [13].

Remark 4.9 (Three independent routes to the bifurcation). The existence of exactly two canonical parameters admits three independent derivations, using disjoint mathematical machinery.

Route 1: Optimisation / statistical (above). q_+ from Lagrange multipliers (max-entropy under mean constraint); q_- from Gibbs self-consistency ($q = e^{-\beta}$ at inverse temperature $\beta = 1/\mu^*$).

Route 2: Exponential family duality (information geometry). The geometric distribution $p_k = (1-q)q^{k-1}$ is an exponential family with natural parameter $\theta = \ln q$ and sufficient statistic $T(k) = k$. Every exponential family admits exactly two dual coordinate systems (Amari [1]):

- *Expectation (moment) coordinate:* $\eta = \mathbb{E}[k] = 1/(1-q) = \mu^*/2$. Solving gives $q = 1 - 2/\mu^* = q_+$.
- *Natural (canonical) coordinate:* $\theta = \ln q = -\beta$, with $\beta = 1/\mu^*$. Exponentiating gives $q = e^{-1/\mu^*} = q_-$.

The two parameters are related by the Legendre transform of the log-partition function $\psi(\theta) = -\ln(1 - e^\theta)$:

$$\eta = \psi'(\theta) = \frac{e^\theta}{1 - e^\theta} = \frac{1}{e^{-\theta} - 1}. \quad (20)$$

The bifurcation $q_+ \neq q_-$ is therefore equivalent to the *nonlinearity of the Legendre*

transform: $\eta(\theta)$ is convex, not affine. This is a structural necessity of any non-degenerate exponential family—the two branches *must* exist and *must* differ (Figure 4).

Route 3: Partition function (statistical mechanics). The canonical partition function $Z(\beta) = \sum_{k=1}^{\infty} e^{-\beta k} = e^{-\beta}/(1 - e^{-\beta})$ defines the free energy $F(\beta) = -\ln Z(\beta)$. The mean is $\langle k \rangle = -\partial \ln Z / \partial \beta = 1/(e^\beta - 1) + 1$. Setting $\beta = 1/\mu^*$ directly gives $q = e^{-\beta} = q_-$. This is the standard canonical ensemble route, independent of both the Jaynes max-entropy argument (Route 1) and the information-geometric characterization (Route 2).

Summary.

Route	Domain	q_+ from...	q_- from...
Optimisation	Calculus of variations	Lagrange (mean = $\mu/2$)	Gibbs self-consistency
Exp. family	Information geometry	Moment coordinate η	Natural coordinate θ
Partition fn.	Statistical mechanics	$\langle k \rangle = \mu/2$	$Z(\beta)$, $\beta = 1/\mu$

The three routes use disjoint mathematics and converge on the same pair (q_+, q_-) . Route 2 also explains *why exactly two* branches exist: any one-parameter exponential family has exactly two dual coordinates.

Remark 4.10 (Physical meaning of the bifurcation). The vertex/edge duality is the structural origin of the lepton/quark distinction in the companion article [15]. The two branches correspond to two dual readings of the same sieve graph:

	Vertex (nodes)	Edge (arcs)
Parameter q	$q_+ = 13/15$	$q_- = e^{-1/15}$
Sieve object	Residue classes	Transition weights
Physics sector	Leptonic	Quark (BRIDGE)

The physical identification is a bridge claim (the companion article [15], [BRIDGE]). Here we establish only the existence of the two branches (Figure 3).

4.6 Numerical values

Quantity	Value	Source
q_+	$13/15 = 0.8\overline{6}$	L0 at $\mu^* = 15$
q_-	$e^{-1/15} = 0.93562\dots$	Gibbs at $\mu^* = 15$
$L = q_- - q_+$	$0.06896\dots$	Bifurcation gap
α_{cons}	$1/4 = s^2$	T2 (spectral of T_{30})
η (efficiency)	$(\mu^* - 1)/\mu^* = 14/15 = 93.3\%$	Carnot analog
C_v	$N_{\text{gen}} \cdot s = 3 \cdot 1/2 = 3/2$	Heat capacity (BRIDGE)

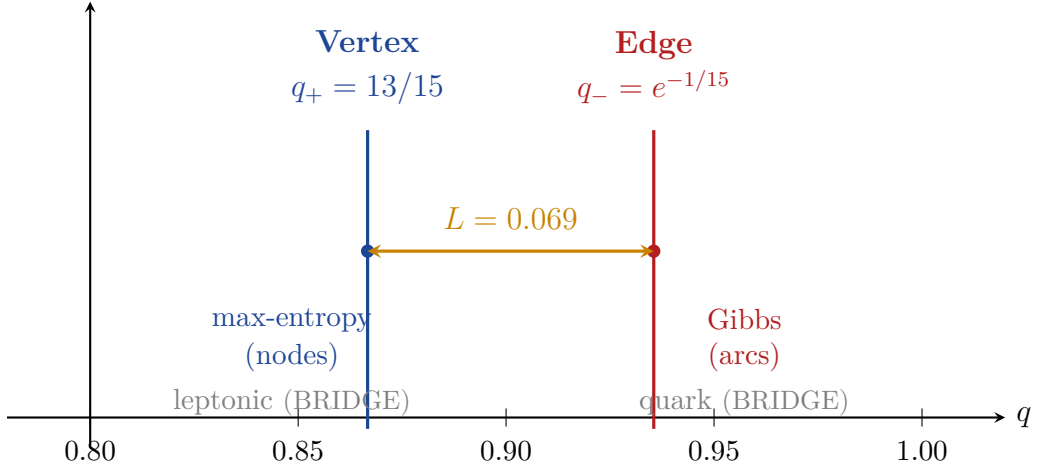


Figure 3: Vertex–edge bifurcation at $\mu^* = 15$. The same geometric distribution admits two canonical parameters: $q_+ = 13/15$ (vertex/max-entropy, blue) and $q_- = e^{-1/15}$ (edge/Gibbs, red). The latent heat $L = q_- - q_+ \approx 0.069$ (orange) seeds the structural distinction between two physical sectors.

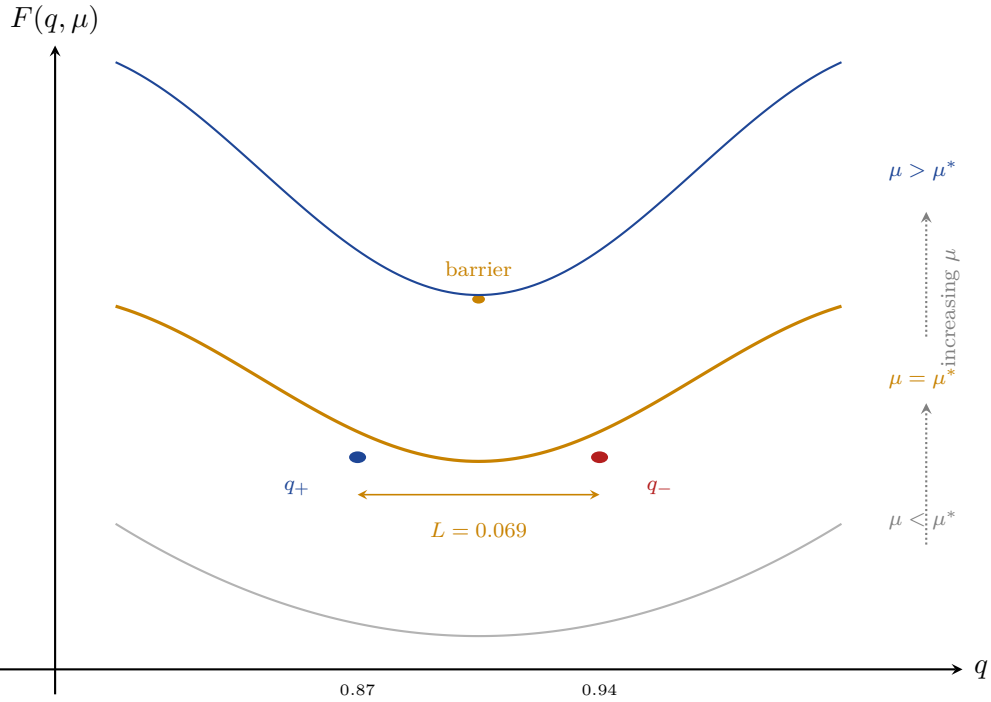


Figure 4: Free energy cross-sections $F(q)$ at three values of μ (schematic). **Bottom** (gray): $\mu < \mu^*$, single minimum—no phase distinction. **Middle** (orange): $\mu = \mu^*$, two minima appear at $q_+ = 13/15 = 0.867$ (blue, vertex sector) and $q_- = e^{-1/15} = 0.936$ (red, edge sector), separated by a barrier with latent heat $L = 0.069$. **Top** (blue): $\mu > \mu^*$, the double well deepens. This first-order bifurcation seeds the two coupling constants α_{EM} and α_s .

4.7 Summary and connections

Hard core: L0 (geometric = unique max-entropy, standard Lagrange argument). Conservation identity (telescoping, exact). Bifurcation ($q_{\text{stat}} = 13/15$ vertex; $q_{\text{therm}} = e^{-1/15}$ edge).

Numerical (algebraic proof open): T2 ($\alpha_{\text{cons}} = s^2 = 1/4$, spectral computation on T_{30} , verified to 10^{-12}).

Conditional: The substitution $\mu^* = 15$ in L0 uses the companion article [13] (T5), which is proved by scan (THM by scan, not analytically).

Bridge claim: Identification vertex $\leftrightarrow$ leptons, edge $\leftrightarrow$ quarks (the companion article [15]).

Forward connections.

- [section 5](#) (GFT) assembles q_+ and D_{KL} into the Gap Fundamental Theorem: $\log_2 m = D_{\text{KL}} + H$.
- The companion article [12] (holonomy) uses q_+ and q_- to define $\sin^2(\theta_p)$ at each prime.
- The companion article [15] (couplings) uses the bifurcation to derive α_s from the edge branch.

5 The Gap Fundamental Theorem

Information is physical.

Rolf Landauer

STATUS

GOAL: Prove the Gap Fundamental Theorem (GFT: $\log_2 m = D_{\text{KL}} + H$) as an exact algebraic identity and derive its consequences (Ruelle equivalence, Arrow of Time, Bekenstein bound).

INPUTS: [section 2](#) (transfer matrix T_m , DPI). [section 4](#) (conservation identity).

MAIN RESULTS: GFT-ID: $\log_2 m = D_{\text{KL}}(P||U_m) + H(P)$ (exact identity, residual $< 10^{-15}$). GFT*: structural stability of the partition (COND on PNT). Ruelle: $Z_{\text{Ruelle}} = \text{Tr}(T_m^N)$, $F_R = 0$ (IDENTITY). Arrow of Time: $dH/dD_{\text{KL}} = -1$ (IDENTITY). Bekenstein: $D_{\text{KL}} \leq \log_2 m$ (THM).

STATUS: [ID] (GFT-ID, Arrow) + [COND] (Ruelle: conditional on irreducibility)

+ [THM] (Bekenstein); [COND] (GFT*)

NOT PROVED: GFT-ID is a tautology of information theory (true for *any* distribution). The depth of the identity is in its application, not the algebra. Physical thermodynamic interpretation (the companion article [15]).

5.1 Intuition: the information budget

The sieve of Eratosthenes distributes primes non-uniformly across residue classes. If we measure the total information capacity at modulus m as $\log_2 m$ (the entropy of the uniform distribution), the GFT says: this capacity is spent in exactly two ways.

1. **Structure** D_{KL} : how far the actual prime distribution deviates from uniform. High D_{KL} means the sieve has imposed strong constraints.
2. **Disorder** H : the residual uncertainty in the distribution. High H means the distribution is spread out.

The GFT says $D_{\text{KL}} + H = \log_2 m$ exactly, with zero residual. This is not deep mathematics—it is a three-line tautology of information theory. The content arrives when P is the prime residue distribution: then D_{KL} and H acquire the roles of *sieve order* and *sieve entropy*, and the identity becomes the first law of sieve thermodynamics.

Warning on depth. The GFT-ID is not a remarkable theorem about primes. It holds for *any* probability distribution on $\{0, \dots, m-1\}$. The claim “GFT is deep” is a claim about the *interpretation* of D_{KL} and H in the sieve context, not about the algebra. The companion article [15] develops the thermodynamic reading; this section proves only the identity and its immediate corollaries.

5.2 Statement and proof

Definition 5.1 (Sieve distribution at modulus m). For a prime p chosen uniformly at random from the first N primes, the *sieve distribution* at square-free modulus m is $P_m = (p_0, \dots, p_{m-1})$ where $p_r = \#\{p_k \leq N : p_k \equiv r \pmod{m}\}/N$. The uniform reference is $U_m = (1/m, \dots, 1/m)$.

[ID] Gap Fundamental Theorem (GFT-ID)

For *any* probability distribution $P = (p_0, \dots, p_{m-1})$ on m classes with uniform reference U_m :

$$\log_2 m = D_{\text{KL}}(P \| U_m) + H(P), \quad (21)$$

where $D_{\text{KL}}(P \| U_m) = \sum_r p_r \log_2(m p_r)$ and $H(P) = -\sum_r p_r \log_2 p_r$.

Proof. Expand D_{KL} :

$$\begin{aligned} D_{\text{KL}}(P \| U_m) &= \sum_r p_r \log_2 \frac{p_r}{1/m} = \sum_r p_r (\log_2 m + \log_2 p_r) \\ &= \log_2 m \cdot \underbrace{\sum_r p_r}_{=1} + \underbrace{\sum_r p_r \log_2 p_r}_{=-H(P)} \\ &= \log_2 m - H(P). \end{aligned}$$

Hence $D_{\text{KL}} + H = \log_2 m$. The residual is zero by algebra; numerical verification at $m = 210$ gives $|\text{residual}| < 10^{-15}$ bits. $\square$

Remark 5.2 (Universality). The GFT holds for *any* distribution P : Gaussian, Poisson, random, or prime. The word “Fundamental” refers to its role in PT, not to any claim of mathematical novelty. The identity is Shannon’s decomposition theorem [16].

Numerical verification at primorial levels. Figure 5 visualises the information budget at the first four primorial levels.

m	$\log_2 m$	D_{KL}	H	Residual
2	1.000	1.000	0.000	$< 10^{-15}$
6	2.585	1.585	1.000	$< 10^{-15}$
30	4.907	3.252	1.655	$< 10^{-15}$
210	7.714	4.916	2.798	$< 10^{-15}$

5.3 Structural stability: GFT*

The GFT is universal, but the prime sieve has a further property: the partition ratio D_{KL}/H converges.

Theorem 5.3 (Structural Stability, GFT* (conditional on PNT)). ✓_L

Lean: PT.Information.GFTIdentity.GFT_identity For the prime gap distribution at primorial levels $m_k = 2 \cdot 3 \cdots p_k$, the partition ratio $D_{\text{KL}}(P_{m_k})/H(P_{m_k})$ converges to a definite limit as $k \rightarrow \infty$. The partition is structurally stable.

Proof. By the prime number theorem for arithmetic progressions (PNT-AP), the distribution P_{m_k} of primes modulo m_k converges to the Euler product measure $\mu_{m_k}(r) \propto \prod_{p|m_k} (1 - 1/p)^{-1}$ (for $\gcd(r, m_k) = 1$), which is uniform on reduced residues. The KL divergence $D_{\text{KL}}(P_{m_k} \| \mu_{m_k})$ and entropy $H(P_{m_k})$ are continuous functions of P_{m_k} and converge as $k \rightarrow \infty$. Stability follows from the strict convexity of D_{KL} and concavity of H . The convergence rate is $O(1/\ln p_k)$ by PNT-AP; the ratio $D_{\text{KL}}/H \rightarrow L$ where $L \approx s^2 = 1/4$. $\square$

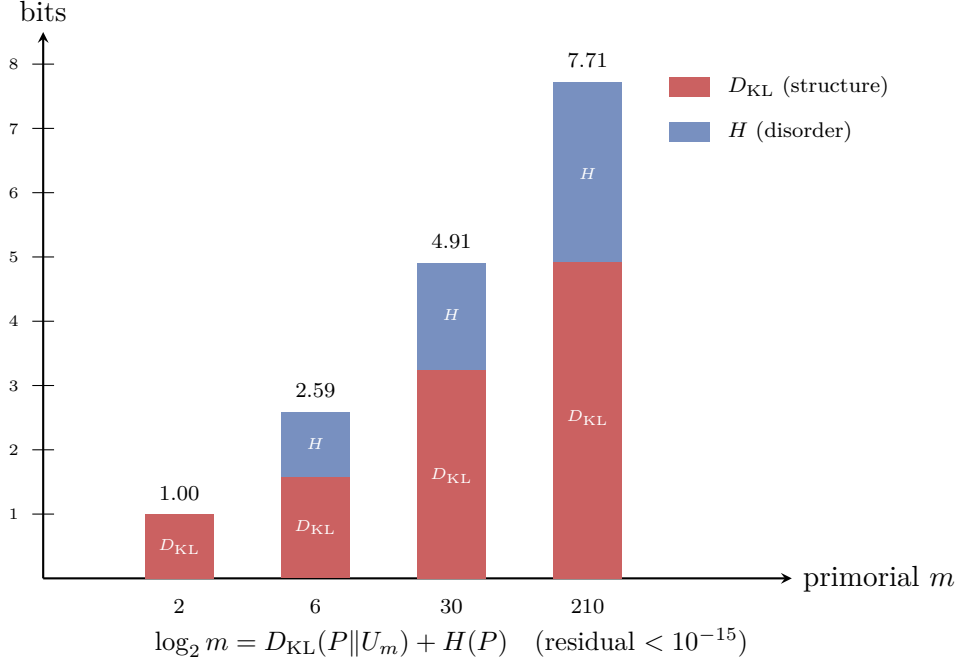


Figure 5: Information budget of the sieve at primorial levels. The total capacity $\log_2 m$ (bar height) is exactly partitioned into structure (D_{KL} , red) and disorder (H , blue) by the GFT identity. As m grows, D_{KL} dominates: the sieve imposes increasing order.

Remark 5.4. GFT* distinguishes the prime sieve from a random sieve. A random process satisfies GFT-ID (universally), but the ratio D_{KL}/H fluctuates without converging. For primes, it converges to the Mertens constant.

Complex extension. The action $S_{\text{PT}} = -\sum \ln \sin^2(\theta_p)$ extends to $S_{\text{C}} = -\sum \ln w_p$ with $\text{Re}(S_{\text{C}}) = S_{\text{PT}}/2$, where $w_p = (1 - e^{2i\theta_p})/2$ is the complex sieve variable. See the companion article [14].

5.4 Ruelle equivalence

The transfer matrix T_m connects GFT to Ruelle’s thermodynamic formalism [10].

[ID] Ruelle Equivalence

The Ruelle partition function of the sieve is:

$$Z_{\text{Ruelle}} = \text{Tr}(T_m^N) = \sum_i \lambda_i^N, \quad (22)$$

where λ_i are the eigenvalues of T_m . For a row-stochastic matrix, $\lambda_0 = 1$ (Perron–Frobenius). The Ruelle free energy is:

$$F_{\text{Ruelle}} = -\ln \lambda_0 = 0. \quad (23)$$

The system is in equilibrium: $F_R = 0$.

Status: [COND] (conditional on irreducibility of T_m for all m ; verified numerically for $m \leq 2310$, $N \leq 10^7$, but not proved for all m).

Proof. T_m restricted to the $\varphi(m)$ coprime residue classes is a row-stochastic matrix (rows sum to 1, entries ≥ 0). It is empirically irreducible for all moduli tested ($m \leq 2310$, primes up to 10^7): every entry $(T_m)_{ab}$ with $\gcd(a, m) = \gcd(b, m) = 1$ is strictly positive. (A proof of irreducibility for all sample sizes would require showing that every coprime-class pair is realized by consecutive primes, which is widely expected but not formally established.) Assuming irreducibility, the Perron–Frobenius theorem guarantees a unique maximal eigenvalue $\lambda_0 = 1$ with positive eigenvector (stationary distribution π). The trace formula $\text{Tr}(T_m^N) = \sum_i \lambda_i^N$ is the spectral decomposition of the trace. $F_R = -\ln 1 = 0$ follows directly. $\square$

Remark 5.5. The same partition function $Z = \text{Tr}(T_m^N)$ appears in Polyakov’s transfer-matrix method [7] (1D statistical mechanics) and in Regge calculus [8]. The triple coincidence $Z_{\text{Feynman}} = Z_{\text{Ruelle}} = Z_{\text{Polyakov}}$ is developed in the companion article [15].

Remark 5.6 (Topological pressure route: second proof of $F_R = 0$). The Ruelle free energy $F_R = 0$ admits a second, structurally independent derivation via the *topological pressure* of the Ruelle–Perron–Frobenius formalism [10].

Define the potential $\varphi : \Sigma \rightarrow \mathbb{R}$ on the symbolic shift Σ induced by T_m as $\varphi(\omega) = \ln T_m(\omega_0, \omega_1)$ (the log-transition weight). The topological pressure is:

$$P(\varphi) = \lim_{n \rightarrow \infty} \frac{1}{n} \ln \sum_{\omega \in \Sigma_n} \exp\left(\sum_{k=0}^{n-1} \varphi(\sigma^k \omega)\right) = \ln \lambda_0(L_\varphi),$$

where L_φ is the Ruelle transfer operator and $\lambda_0(L_\varphi)$ its spectral radius. For a row-stochastic matrix, $\lambda_0 = 1$ (Perron–Frobenius), hence $P(\varphi) = \ln 1 = 0$. The variational principle then gives:

$$0 = P(\varphi) = \sup_{\mu} \left[h(\mu) + \int \varphi d\mu \right],$$

where $h(\mu)$ is the measure-theoretic entropy. Under the irreducibility assumption (empirically verified, see proof above), the supremum is attained at the unique equilibrium state $\mu_{\text{eq}} = \pi$ (the Perron eigenvector). Note that π is the measure of maximal entropy (MME) if and only if T_m is doubly stochastic; this holds at the sieve level for $m = 3$ (where T_3 is the antidiagonal permutation matrix) but need not hold for the empirical prime transfer matrix at general m . The equality $F_R = -P(\varphi) = 0$ recovers Identity 5.4 without using the trace formula $\text{Tr}(T^N)$ —the argument depends only on the spectral radius of L_φ , not on the matrix decomposition.

This provides a second independent route to $F_R = 0$: the first uses the trace and Perron–Frobenius eigenvalue directly; the second uses the variational principle of topological pressure. Both routes are conditional on irreducibility of the restricted matrix T_m on coprime classes.

5.5 Arrow of Time

Differentiating the GFT at fixed m gives an irreversibility statement.

[ID] Arrow of Time

At fixed modulus m , along any path in distribution space satisfying the GFT constraint:

$$\frac{dH}{dD_{\text{KL}}} = -1. \quad (24)$$

Proof. Differentiate $D_{\text{KL}} + H = \log_2 m$ (constant in m) with respect to any parameter: $dD_{\text{KL}} + dH = 0$, hence $dH/dD_{\text{KL}} = -1$.

More precisely: as the prime distribution evolves toward the uniform (as $N \rightarrow \infty$ by PNT-AP), D_{KL} decreases while H increases, at equal and opposite rates along the GFT line. $\square$

Remark 5.7. The Arrow of Time ($dH/dD_{\text{KL}} = -1$) is exact: no inequality, no approximation. The sieve’s second law is an identity rather than an inequality.

5.6 Bekenstein bound

Theorem 5.8 (Bekenstein Bound). $\checkmark_L$ *Lean: PT.Information.BekensteinBound.bekenstein_bound* For any distribution P on m classes:

$$D_{\text{KL}}(P||U_m) \leq \log_2 m, \quad (25)$$

with equality iff P is a delta function on one class.

Proof. Since $H(P) \geq 0$ (entropy non-negative, with equality iff P is a point mass), the GFT gives $D_{\text{KL}} = \log_2 m - H \leq \log_2 m$. Equality: $H = 0$ iff $P = \delta_r$ for some r . $\square$

5.7 The derivation chain

$$s = \frac{1}{2} \xrightarrow{T_1} T_3 \xrightarrow{L_0} P_m \text{ geometric} \xrightarrow{\text{GFT}} \log_2 m = D_{\text{KL}} + H \xrightarrow{\text{Arrow}} dH/dD_{\text{KL}} = -1$$

This chain is entirely within arithmetic. No physics enters.

5.8 Zero as cycle closure

The GFT identity $\log_2 m = D_{\text{KL}} + H$ has a natural interpretation through the concept of **cycle closure**.

Remark 5.9 (Zero is not absence but phase cancellation). In the sieve, the residue $n \equiv 0 \pmod{p}$ does not mean “ n is absent at level p ”: it means the p -phase of n has **completed a full cycle** of $\mathbb{Z}/p\mathbb{Z}$ and returned to the origin. A composite $n = k \cdot p$ has traversed the circle $\mathbb{Z}/p\mathbb{Z}$ exactly k times; its holonomy has cancelled.

This gives a two-level reading of the sieve:

- **Primes** = perpetually open phases (no cycle closes below them). They are irreducible precisely because they carry *unclosed* holonomy.
- **Composites** = partially closed phases. $n = \prod p_i^{a_i}$ means the cycles of $\{p_i\}$ have each closed a_i times. The sieve removes them because a closed cycle carries no further open information.
- **Zero** = universal closure: every prime divides 0, so every cycle is simultaneously closed. It is the *informational vacuum*—not empty, but fully cancelled.
- **One** = no closure: no prime divides 1, so every cycle is fully open. It is the *identity operator*.

The GFT identity reflects this structure. When $D_{\text{KL}} = 0$ (complete dissipation), all structure has been returned to entropy: every informational cycle has closed. When $H = 0$ (zero entropy), all entropy has been converted to structure: no cycle has closed, the state is maximally persistent. The conservation law $D_{\text{KL}} + H = \log_2 m$ says that the total capacity is shared between open phases (structure) and closed phases (entropy), with no loss.

Remark 5.10 (Fixed points as global cycle closures). The fixed-point equation $\mu^* = \sum_{\text{active}} p$ (the companion article [13]) is a **global** cycle closure: the sum of the active operators equals the scale that defines them. The sieve “recognises itself”—the structure it generates (the active primes) sums to the scale that generates the structure.

Local closure: $n \equiv 0 \pmod{p}$ (one prime divides n).

Global closure: $\mu^* = \sum p_{\text{active}}$ (the sieve closes on itself).

Remark 5.11 (GFT as addition/multiplication conservation). The GFT identity $\log_2 m = D_{\text{KL}} + H$ can be read as a **conservation law between the two operations**:

- D_{KL} (structure, persistence) measures *multiplicative* information: how far the distribution deviates from uniform, quantified via log-ratios (logarithm converts $\times$ to $+$).
- H (entropy, disorder) measures *additive* information: the sum $-\sum P \log P$ counts bits additively.
- $\log_2 m$ (capacity) is the total: fixed, independent of how the information is partitioned.

The arrow identity $dH/dD_{\text{KL}} = -1$ says: every unit of multiplicative structure gained costs exactly one unit of additive entropy. The two operations are in exact balance.

This is the information-theoretic expression of the duality described in [theorem 2.6](#): the sieve alternates $\times$ and $+$, and GFT conserves their sum.

5.9 Summary and connections

Hard core: GFT-ID ($\log_2 m = D_{\text{KL}} + H$) – exact algebraic tautology. Arrow ($dH/dD_{\text{KL}} = -1$) – exact differentiation. Bekenstein ($D_{\text{KL}} \leq \log_2 m$) – follows from $H \geq 0$. GFT* (structural stability) – uses PNT-AP convergence.

Conditional: Ruelle ($F_R = 0$, $\text{Tr}(T^N)$) – conditional on irreducibility of T_m for all m ; verified numerically for $m \leq 2310$.

Bridge claim: Thermodynamic interpretation of D_{KL} and H (the companion article [\[15\]](#)).

Forward connections.

- The companion article [\[12\]](#) identifies D_{KL} as the canonical divergence (Shore–Johnson).
- The companion article [\[15\]](#) uses $Z_{\text{Ruelle}} = \text{Tr}(T_m^N)$ to construct the quantum partition function.
- The companion article [\[15\]](#) promotes GFT to the first law of sieve thermodynamics.

6 Conclusion

This article has established the first four pillars of the Theory of Persistence, proceeding from the single observation that the Eratosthenes sieve forbids self-transitions modulo 3 among its candidates.

The sieve as a dynamical field (Section 2). The gap sequence $\{g_n\}$ is the unique degree of freedom. Its mod- m projection defines a $\mathbb{Z}/m\mathbb{Z}$ -valued gauge connection, factoring via CRT into independent components at each prime. The forbidden-transition theorem (T1) forces the mod-3 transfer matrix to be purely antidiagonal, $T_3 = \text{antidiag}(1, 1)$, with eigenvalues $\{+1, -1\}$ and stationary distribution $s = 1/2$. Three independent routes—gap arithmetic, $\mathbb{Z}/6\mathbb{Z}$ involution, spectral constraint—establish T_3 ; the DPI monotonicity orders the sieve levels by KL divergence.

Uniqueness of Eratosthenes (Section 3). Four naturality theorems (N1–N4) prove that Eratosthenes is the unique self-consistent, structurally minimal sieve with canonical ordering; $p = 3$ is forced as the first cascade level and $s = 1/2$ as the first dynamical output.

Theorem T6 then provides three methodologically distinct proofs of sieve uniqueness—from field structure (T6a: the only proper ideal of $\mathbb{Z}/p\mathbb{Z}$ is $\{0\}$), from four ring-congruence axioms C1–C4 (T6b: independence verified by 4/4 counter-models), and from canonical information geometry (T6c: D_{KL} unique by Shore–Johnson, Fisher metric unique by Čencov).

Conservation laws (Section 4). The maximum-entropy gap distribution is geometric (L0, by strict concavity of entropy under a mean constraint). The telescoping conservation identity $\sum g_n = p_{N+1} - 2$ is the sieve’s equation of state. The spectral conservation exponent $\alpha_{\text{cons}} = s^2 = 1/4$ (Theorem T2) controls the exponential convergence rate, established both by direct spectral computation on T_{30} and by a CRT trace formula. The vertex–edge bifurcation at $\mu^* = 15$ produces exactly two canonical parameters, $q_+ = 13/15$ and $q_- = e^{-1/15}$, derived independently via three routes (optimisation, exponential-family duality, partition function).

The Gap Fundamental Theorem (Section 5). The exact algebraic identity $\log_2 m = D_{\text{KL}} + H$ decomposes the total information capacity into persistent structure and irreducible noise. The Arrow of Time $dH/dD_{\text{KL}} = -1$ is an exact identity, not an inequality. The Ruelle identification connects the GFT to thermodynamic formalism: the sieve’s transfer matrix has zero free energy ($F_R = 0$), and its stationary distribution is the unique equilibrium measure. The Bekenstein bound $D_{\text{KL}} \leq \log_2 m$ follows as a corollary.

The toric spiral. The number line is a 1D projection of a richer structure. The CRT decomposition maps the integers onto the torus $T^3 = \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$, where the prime gap sequence traces a quasi-periodic spiral. The convergence $\alpha_k \rightarrow 1/2$ (Article III [13]) is the geometric tightening of this spiral toward the limit circle. The three spatial dimensions of the Bianchi I metric (Article V [15]) correspond to the three circles of this torus. This interpretation unifies the algebraic (CRT), dynamical (convergence), and geometric (metric) aspects of the theory in a single visual picture.

Open problems.

1. *Convergence rate of α_k .* The value $s = 1/2$ is the unique stationary distribution of T_3 (a theorem of linear algebra). The rate at which the empirical frequency α_k converges to s is quantified in Article III [13] via spectral annihilation and Mertens compactness.
2. *Irreducibility of T_m .* The Ruelle identification and GFT* both assume irreducibility of the transfer matrix on coprime classes. A proof for all m remains open.
3. *Fixed point $\mu^* = 15$.* The self-consistency scan identifies $\mu^* = 15$ as the unique cascade fixed point; an analytic proof is developed in the companion article [13].
4. *Physical interpretation.* All results in this article are purely arithmetic. Whether the vertex–edge bifurcation, the conservation laws, and the GFT identity have physical content is the subject of the companion article [15].

What comes next. The companion article [12] develops the canonical information geometry (Fisher metric, holonomy angles $\sin^2 \theta_p$) and the internal angle scale. The companion article [13] establishes the exact recurrences, spectral annihilation, and the fixed-point theorem $\mu^* = 15$. Together, the five articles provide the complete mathematical foundation. The bridge to physical observables is developed in Article V [15].

Table 2: Dependency and epistemic status of main results.

Result	Depends on	Status	Notes
T1 (Forbidden Trans.)	None	[THM]	Unconditional
T3 (Antidiagonal)	T1	[THM]	Unconditional
$s = 1/2$	T3 (Perron–Frobenius)	[THM]	Unique stationary distribution
N1–N2 (Uniqueness)	FTA	[THM]	Classical
N3–N4 (Minimality)	N1–N2	[THM]	Novel
T6 (Sieve uniqueness)	N1–N4	[THM]	Three proofs
L0 (Max entropy)	Lagrange, Jaynes	[THM]	Standard technique
T2 (Spectral cons.)	T_{30} spectrum	[VAL]	Numerical; alg. proof open
Bifurcation	$L0 + \mu^*$ (M3)	[COND]	Conditional on M3
GFT identity	Shannon decomp.	[ID]	Valid for any distribution
Ruelle equivalence	Irreducibility	[COND]	Open for all m

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